

HOL-2770-03-ANS-L

Table of Contents

Secure VCF Infrastructure with VMware vDefend Firewall - (HOL-2770-03-ANS-L) 3

Secure VCF Infrastructure with VMware vDefend Firewall - (HOL-2770-03-ANS-L) 4

Lab Guidance 5

Module 1 - Configure Infrastructure Rules for the VCF Infrastructure 7

Introduction 8

Task 1: Review the lab environment and the security policies that have been created in the NSX Manager 10

Task 2: Enable Infrastructure policies to protect the Management Workload Domain 20

Task 3: Validate the Infrastructure Policies 27

Conclusion 29

Module 2 - Configure Environment Rules for the VCF Infrastructure 30

Introduction 31

Task 1: Review the Environment Security Policies for the VCF Deployment 33

Task 2: Enable Environment policies to protect the VCF Deployment 38

Task 3: Validate the Environment Policies 41

Conclusion 44

Module 3 - Configure Application Rules for the VCF Infrastructure 45

Introduction 46

Task 1: Review the application security policies for the VCF Infrastructure 48

Task 2: Enable application policies to protect the VCF Infrastructure 52

Task 3: Validate the application Policies 56

Conclusion 59

Secure VCF Infrastructure with VMware vDefend Firewall - (HOL-2770-03-ANS-L)

Secure VCF Infrastructure with VMware vDefend Firewall - (HOL-2770-03-ANS-L)

In the past, data center security mostly relied on perimeter protection—"building walls" to keep bad actors out. Today, safeguarding the data center's "east-west" or lateral traffic traversing within applications and data is just as critical. Threat actors are leveraging "Frontier AI" security models to automate and accelerate attacks against private cloud workloads.

Within this lab you will learn how VMware vDefend provides best-of-breed protection with significantly reduced complexity to secure your most critical virtual infrastructure management assets. You will guided through the VMware Validated Solution (VVS) to secure your management workloads using vDefend Distributed Firewall. This lab will focus on VCF 9.1 data center topology and provide you with a complete framework to understanding the lateral security polices that are validated to secure your management domain.

For more information on Lateral Security Design Blueprints please see this blog.

Lab Guidance

Welcome! This lab is available for you to repeat as many times as you want. Use the Table of Contents in the upper right-hand corner of the Lab Manual to jump ahead to any module.

Module	Title	Length	Level
1	Secure VCF Infrastructure with VMware vDefend Firewall	10 min	Beginner
2	Configure Infrastructure Rules for the VCF Infrastructure	15 min	Beginner
3	Configure Environment Rules for the VCF Infrastructure	15 min	Beginner
4	Configure Application Rules for the VCF Infrastructure	15 min	Beginner

Lab Authors:

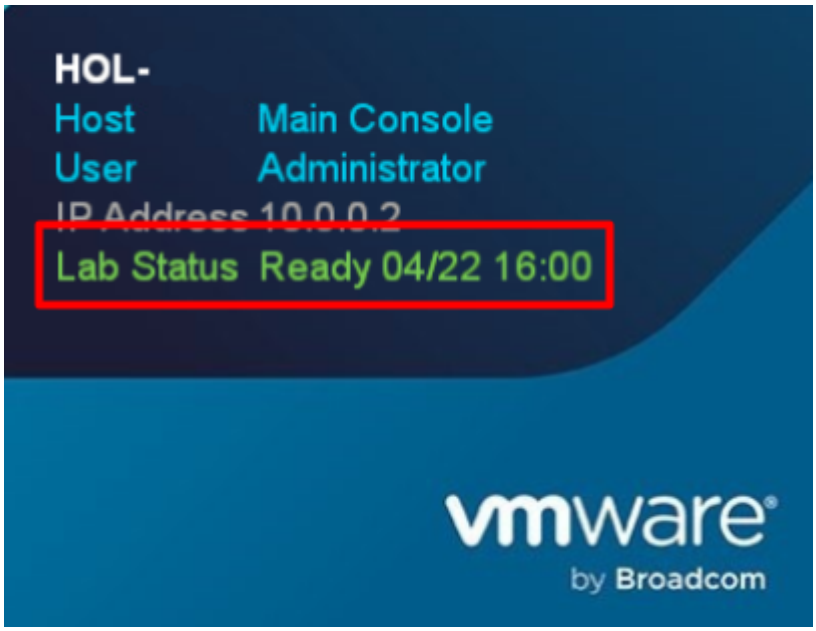
Frank Snyder, Quinton Coelho, Lynne Dark, Joshua Newton, Apoorv Malmane

Special thanks to:

Nikodim Nikodimov, William Stoneman, and Joseph Polcar

First time using Hands-on Labs?

If this is your first time taking a lab you can review the VMware Learning Platform interface before proceeding.



The lab console will indicate when your lab has finished all the startup routines and is ready for you to start. If you see anything other than "Ready", please wait for the status to update. If after 5 minutes your lab has not changed to "Ready", please ask for assistance.

Module 1 - Configure Infrastructure Rules for the VCF Infrastructure

Introduction

Welcome to Module 1!

In this module you will review the current management domain environment and enable the first set of policies to secure the management domain.

Module Overview

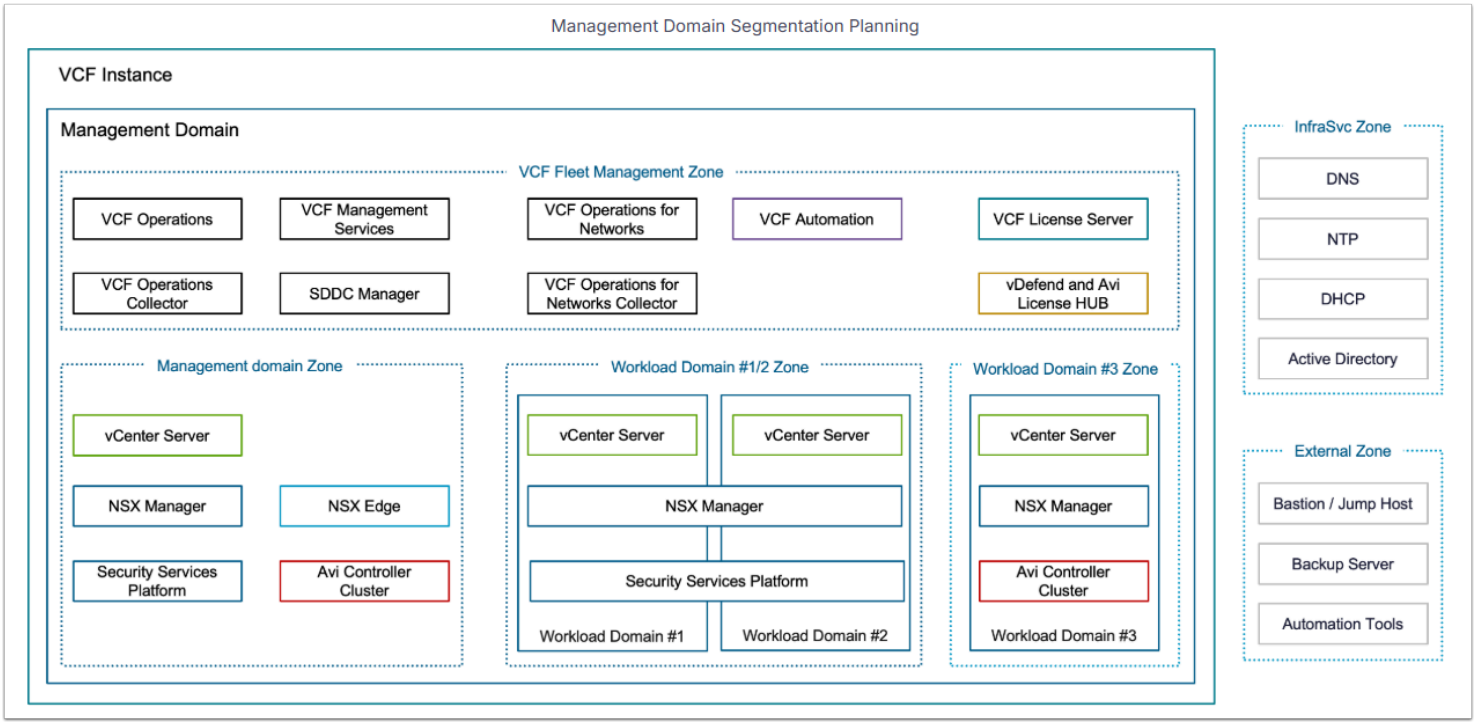
Within this module you will perform the following Tasks to enable and monitor distributed firewall policies for the Management Workload Domain

• Task 1: Review the lab environment and the security policies that have been created in the NSX Manager

• Task 2: Enable Infrastructure policies to protect the Management Workload Domain

• Task 3: Validate the Infrastructure Policies

Management Workload Domain Security Strategy



Before you begin the lab let's take a moment to understand the strategy and terminology that will be used within this lab.

The Management workload domain is broken down into several zones, each of which has a specific purpose and a shared rule set can be applied to each of these zones to describe how they interact with each other.

Zone Definitions:

- **VCF Fleet Management Zone:** This is the zone for all virtual machines that are deployed as part of the VCF Fleet that will communicate with each of the different workload domains deployed within the fleet
- **Management Zone:** This zone contains all hosts (IP based) as well as the management vCenter and NSX Manager with its associated Edges
- **Workload Domain Zones:** A Workload Domain zone is determined by the span of the associated NSX manager
  - Multiple vCenters can be part of a single Workload Domain Zone if they are all connected to the same NSX Manager
  - The zone will contain all vCenters, the associated SSP Platform, the IP addresses of the ESXi Hosts in the associated domains.
  - If an Avi Controller is dedicated to a particular workload domain that would be included with the Workload Domain Zone as well
- **External Zone:** The IP addresses or virtual machines that are allowed to access the Management Domain or will be accessed by the Management Domain for services like backup and external logs
- **InfraSvc Zone:** The IP addresses or virtual machines that provides common services such as DDI, AD, and NTP

Task 1: Review the lab environment and the security policies that have been created in the NSX Manager

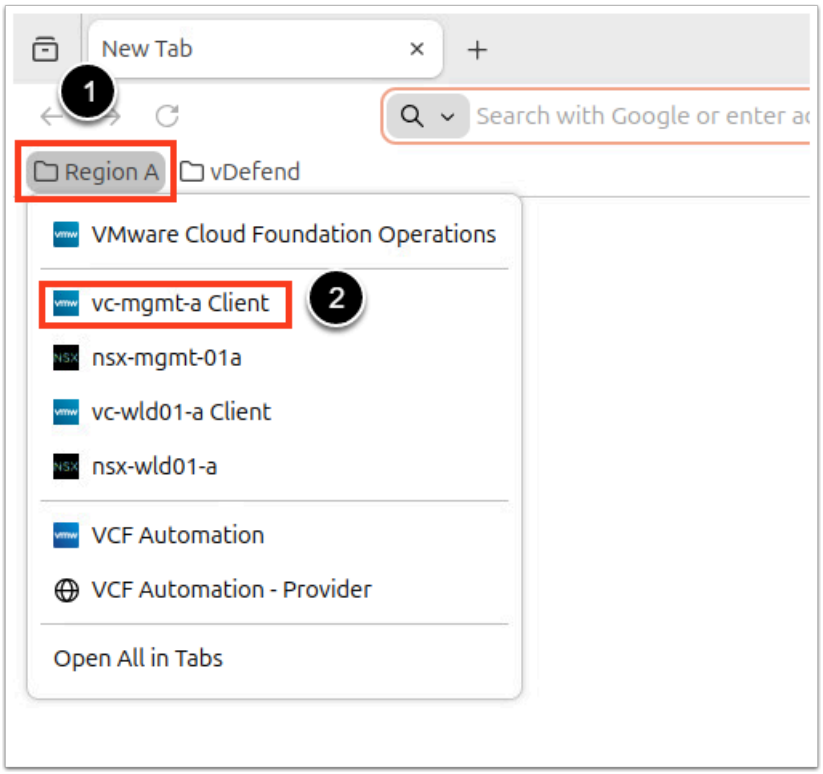
In this task you will access the NSX Manager and Review the Infrastructure policies that have been pre-created for this task.

Open Firefox



1. On the task bar click on the Mozilla Firefox icon

Navigate to vCenter



1. Click on the Region A folder in the Bookmark Toolbar

2. Click on vc-mgmt-a Client

Log into vCenter to review the deployed VMs



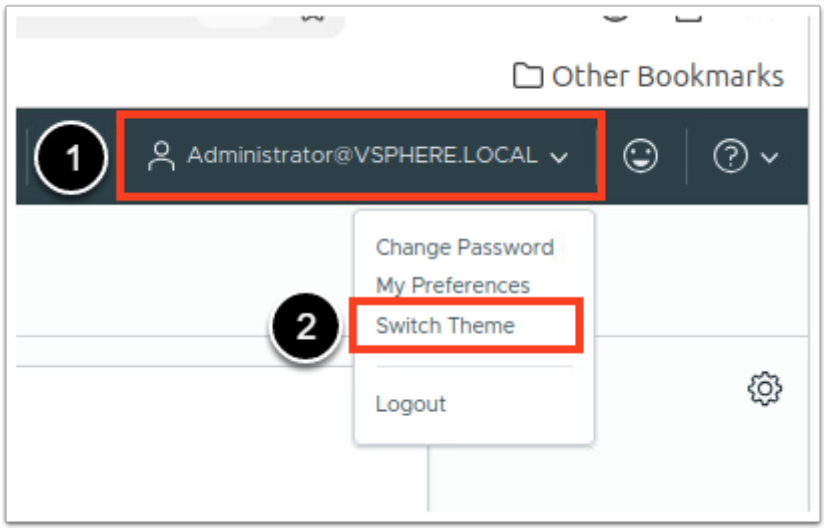
The Username and Password should be populated automatically.

1. Click in the Username field and select administrator@vsphere.local
   This will populate the Username and Password fields
2. Click **LOG IN**

Should the username and password fields be empty or inadvertently removed, please use the following credentials:

- Username: administrator@vsphere.local
- Password: VMware123!VMware123!

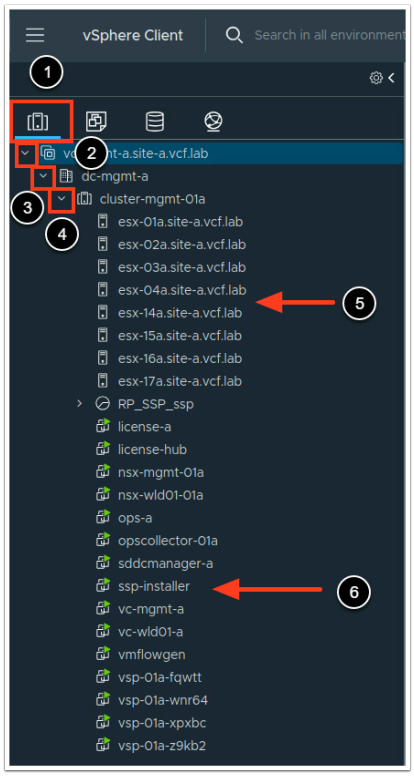
Switch to Dark Mode (Optional)



The default Theme in the lab is light mode. If you prefer dark mode please do the following, if not move on to the next page.

1. Click on the username, **Administrator@VSPHERE.LOCAL,** at the top right.
2. Click on **Switch Theme**

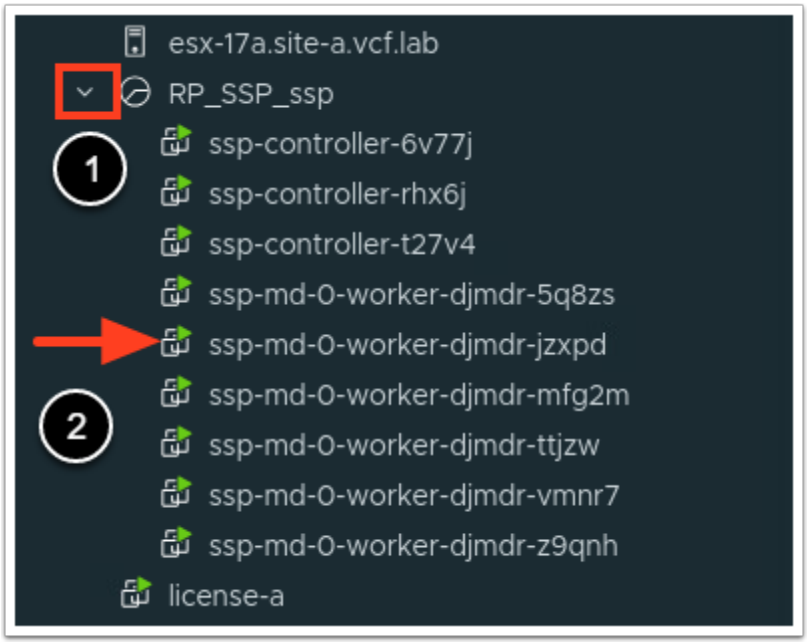
Review VMs that are deployed in the mgmt-a (Management WLD)



After the main page for vCenter is opened you will want to expand the vCenter tree on the left to see what has been deployed:

1. Navigate to the **Hosts and Clusters** view
2. Click on the '>' next to vc-mgmt-a-.site-a.vcf.lab to expand the vCenter level
3. Click on the '>' next to dc-a to expand the data center level
4. Click on the '>' next to cluster-mgmt-01a to expand the cluster level
5. Note the 8 ESXi hosts that are part of this cluster
6. Note the VMs that are deployed in the Management Cluster

Review the SSP VMs

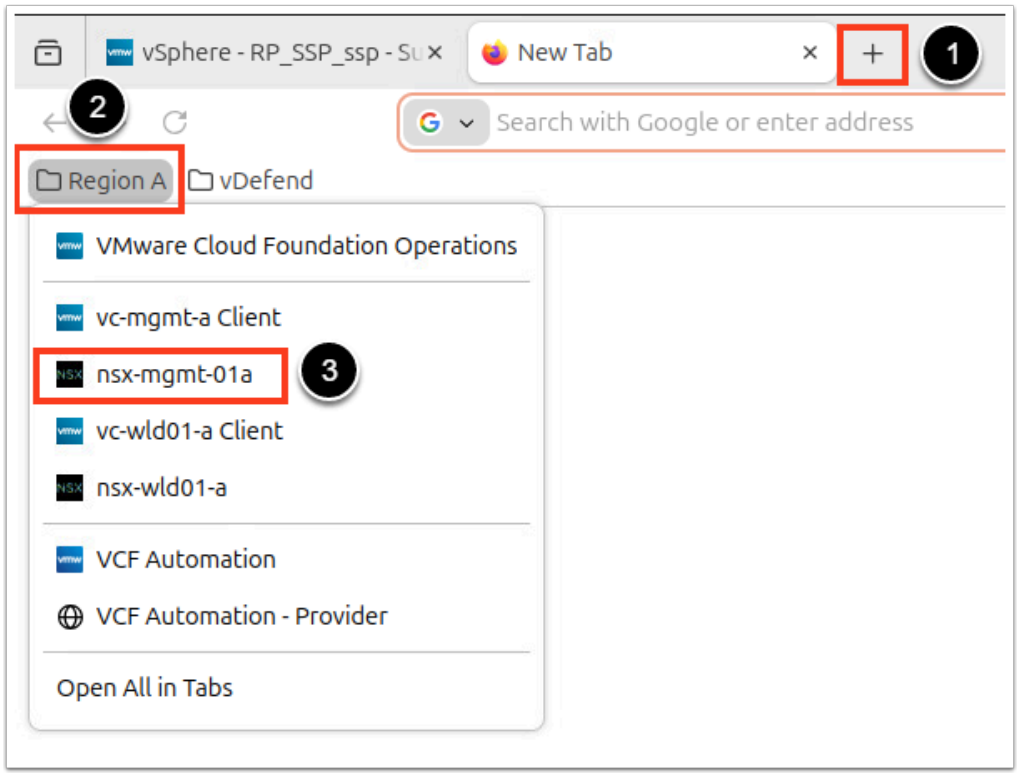


This lab includes a production size vDefend Security Services Platform (SSP) that will be connected to the WLD01 NSX manager. This deployment consists of 3 controller nodes and 6 worker nodes to support the deployment.

1. Click on the '>' next to **RP_SSP_ssp** to show the VMs
2. Take note of the 9 VMs deployed. Here the ssp-controller-* are the 3 Controler nodes and the ssp-md-0-worker-* are the 6 Worker Nodes

Navigate to NSX

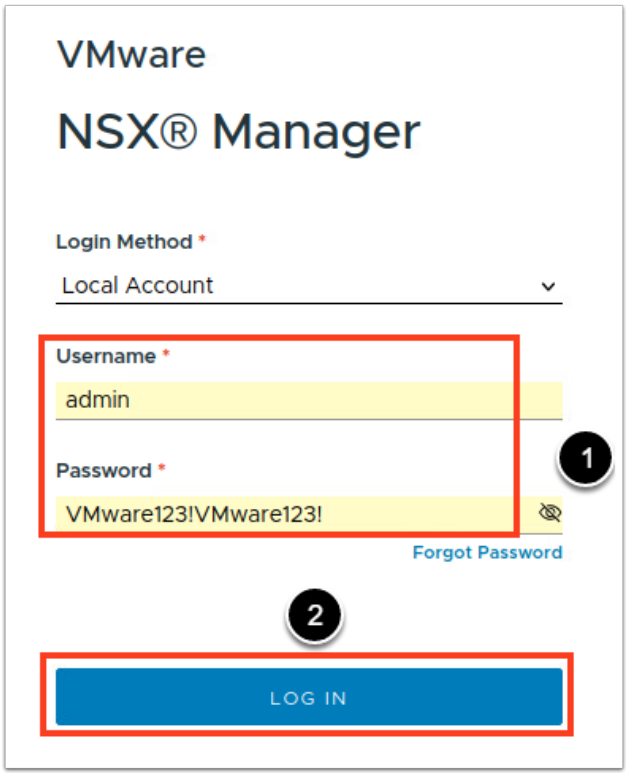
Now that you have reviewed the vCenter infrastructure and deployed VMs it's time to review the NSX configuration.



In a new tab log into the NSX Manager.

1. In Firefox open a new tab by clicking on the '+' icon
2. In the bookmarks Toolbar click on Region A
3. Select nsx-mgmt-01a from the drop down menu

Log Into NSX Manager



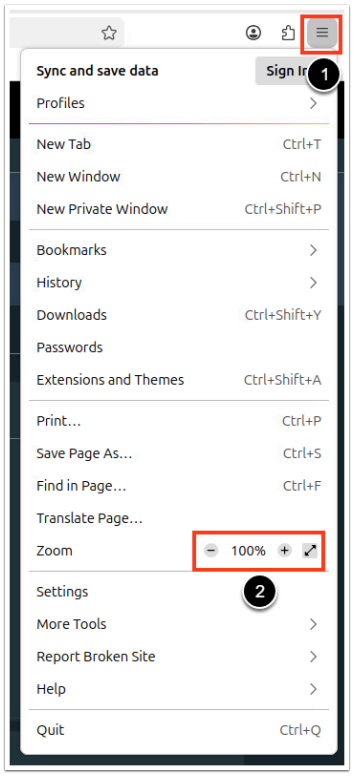
The Username and Password should be populated automatically.

1. Click in the Username field and select admin
   This will populate the the Username and Password fields
2. Click **LOG IN**

> Should the username and password fields be empty or inadvertently removed, please use the following credentials:
>
> - Username: admin
> - Password: VMware123!VMware123!

[Optional but Recommended] Adjust the Browser Zoom

The default Zoom will be set at 80%. In the event the text is either too small or too large you can adjust the browser zoom as you see fit.

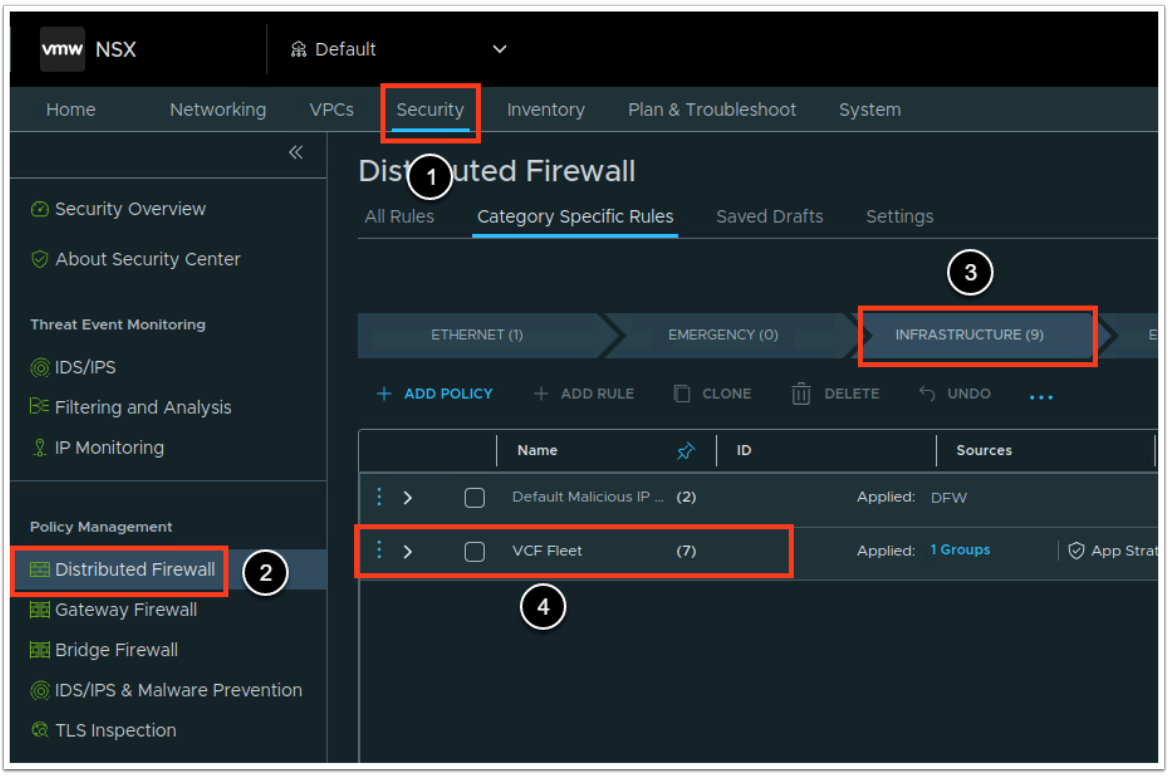


1. On the top right of the browser select the options menu (hamburger icon)
2. Click on the zoom out button '-' to either 90% or 80% or 67% depending on your preference.

Review the Infrastructure Policies to protect the VCF Domain

You may notice some alarms when you first log into the NSX Manager. Please disregard these alarms, they are to be expected.

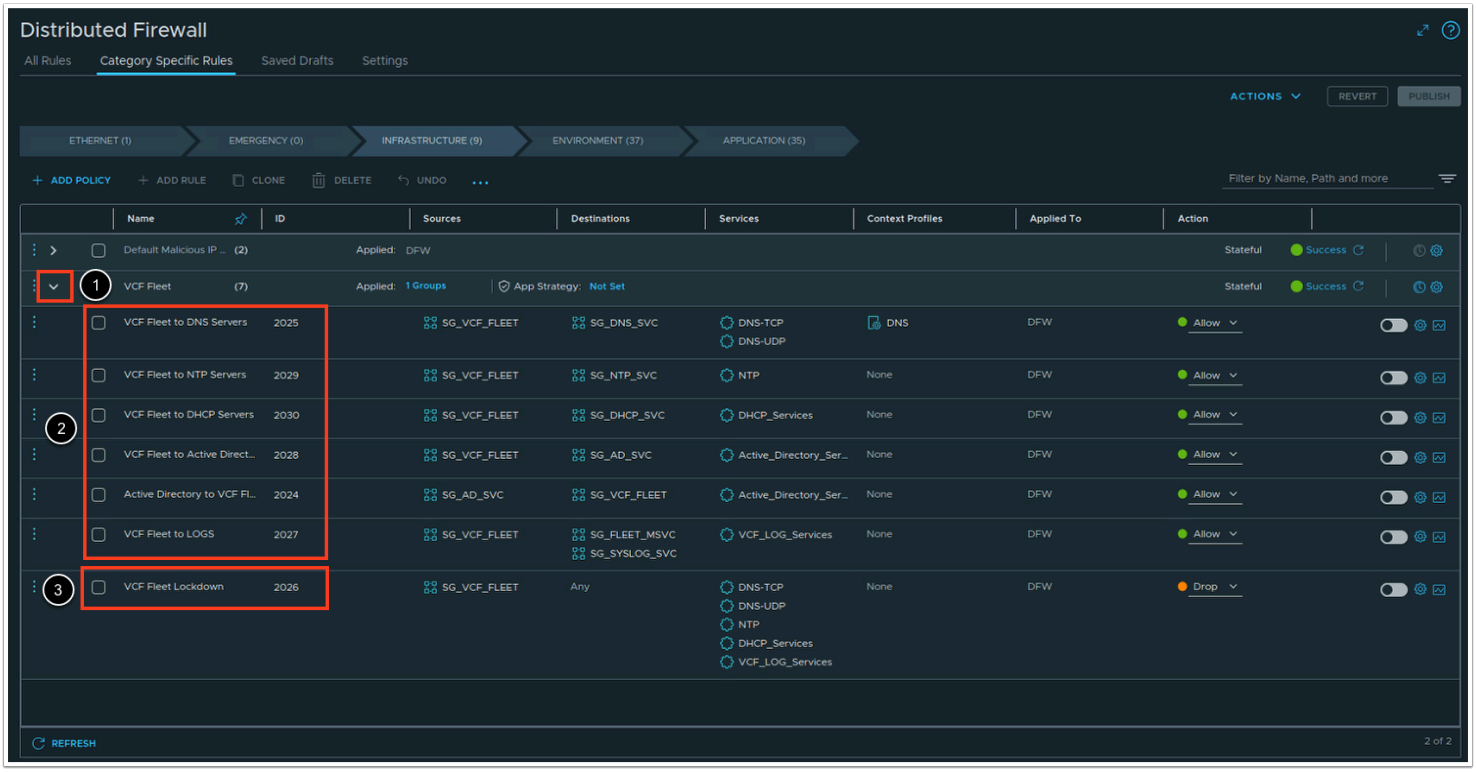
If you see any alerts on the Distributed Firewall page, click on '**Do not show this again**'



The first thing you will be looking at will be the Infrastructure security policies. These policies are implemented to secure infrastructure based protocols such as DNS, DHCP, NTP, AD, and Logging

1. Select **Security** from the top menu
2. Select **Distributed Firewall** from the menu on the left
   - This will bring you to the distributed firewall management section in the NSX Distributed Firewall, by default you will be in the Application Section
3. Navigate to the Infrastructure section by clicking on **INFRASTRUCTURE** in the firewall section menu
4. There is 1 Policy implemented to secure the VCF Fleet with 7 Rules in it.

Review the Infrastructure Rules to protect the VCF Fleet



1. Click on the '>' next to **VCF Fleet** to expand the policy
2. The first 6 rules govern the 5 infrastructure protocols used in the VCF fleet. These are DNS, NTP, DHCP, AD, and Syslog
3. The 7th rule is called a Lockdown rule and this prohibits those 5 protocols detailed above from being used within the VCF Fleet Domain unless explicitly allowed by the first 6 rules.

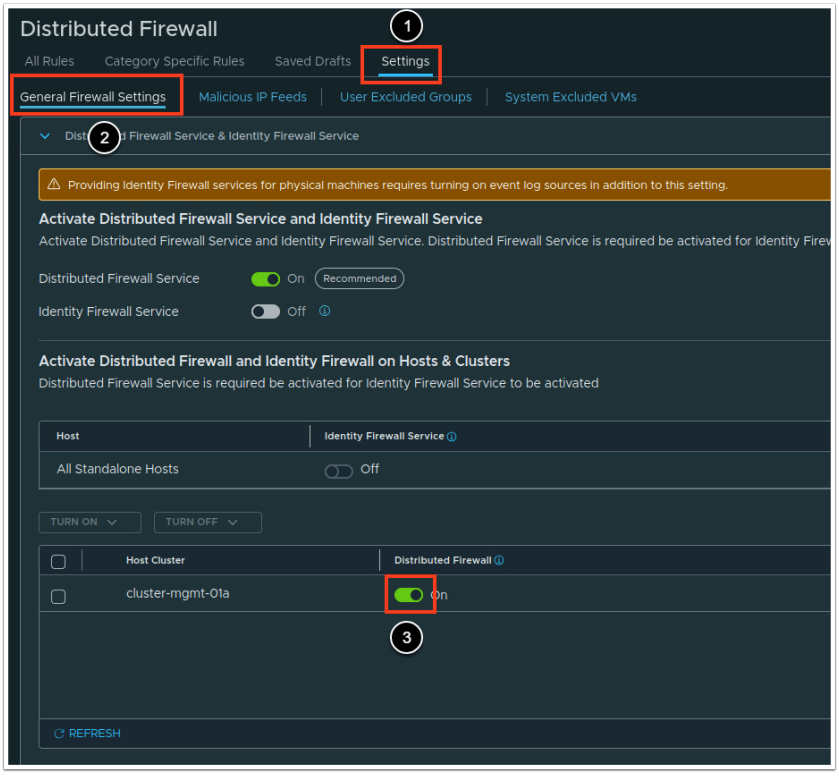


The rules are currently disabled, in the next task you will enable and validate those rules.

Task 2: Enable Infrastructure policies to protect the Management Workload Domain

In this task you will enable the firewall for the management cluster, remove the VCF VMs from the Exclusion List, then enable the rules for the VCF Fleet infrastructure policy.

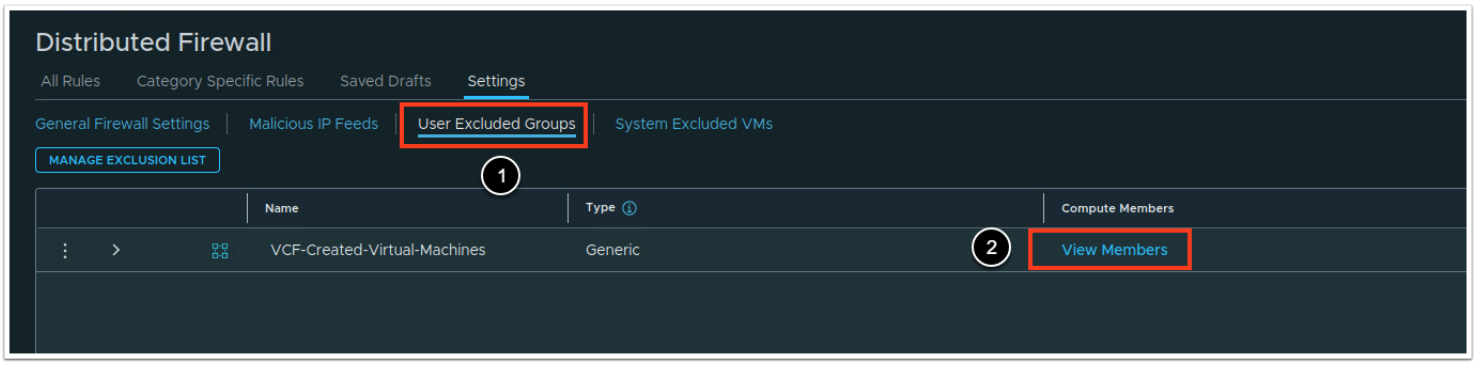
Enable The Distributed Firewall



Before firewall policies can be enforced on any cluster, Distributed Firewall services must be enabled for it.

1. Select **Settings** from the Distributed Firewall menu
2. Select **General Firewall Settings** from the Settings menu
3. Toggle the **Distributed Firewall** status button for cluster-mgmt-01a
4. Click "TURN ON" on the pop-up window

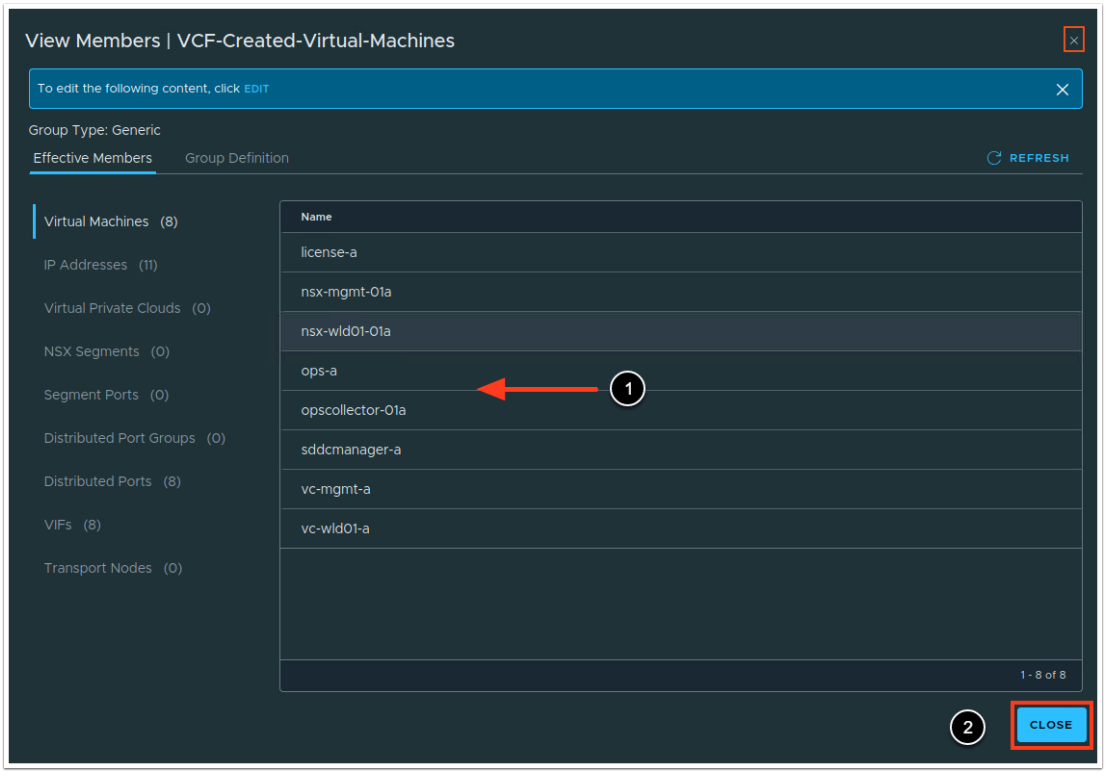
Remove the VMs from the Exclusion List



The Exclusion list is a function that removes VMs from being considered for Distributed Firewall implementation. By Default all VMs created by VCF are added to this list in the group VCF-Created-Virtual-Machines to prevent accidental lockout during initial setup. Before we remove it from the exclusion list lets review what members make up the group.

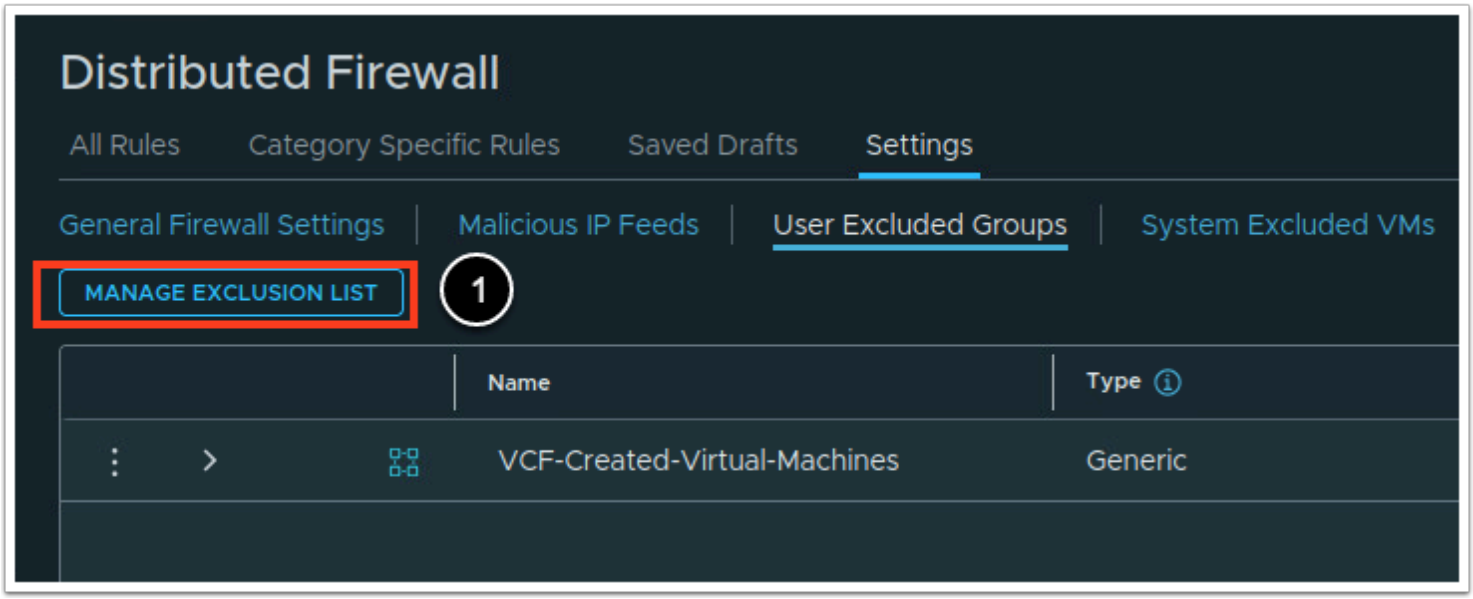
1. Select **User Excluded Groups** from the Settings menu
2. Click on **View Members**

Review the Members of the VCF-Created-Virtual-Machines Group



1. The 8 VMs listed here have all been created by VCF and were statically added to this group.
2. Click **Close**

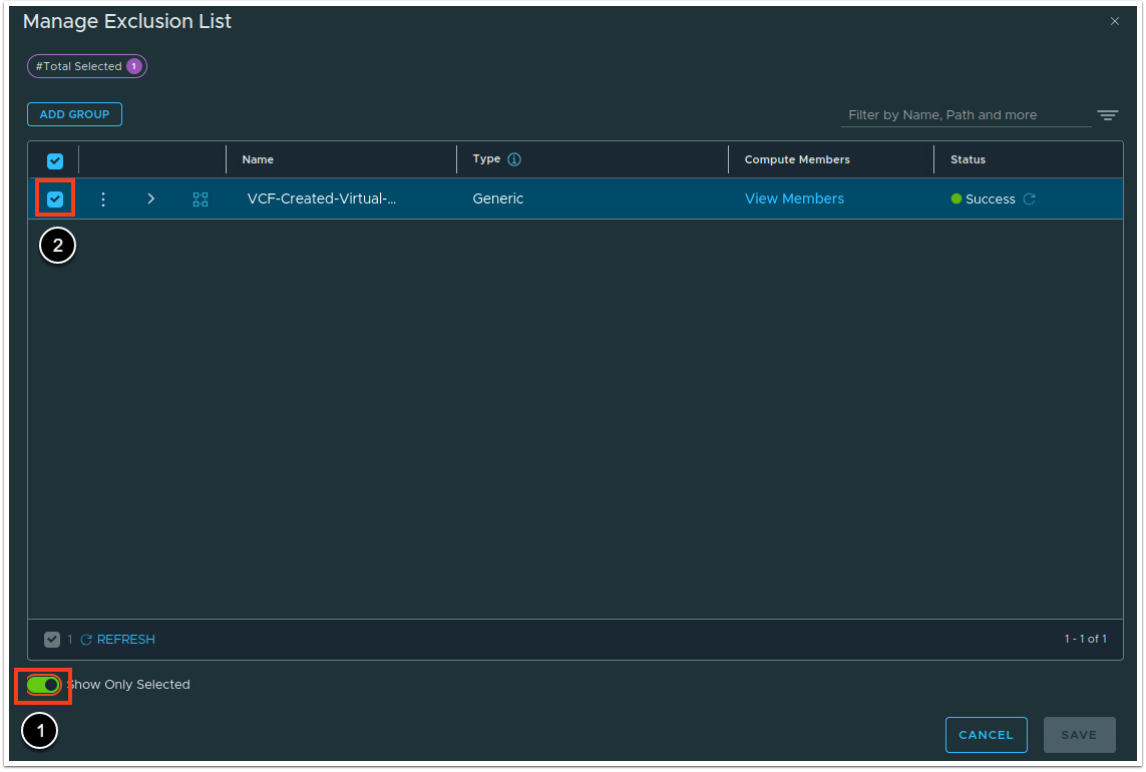
Manage the Exclusion List



To remove those 8 VMs from the exclusion list you will remove the group from the list rather than the VMs from the group.

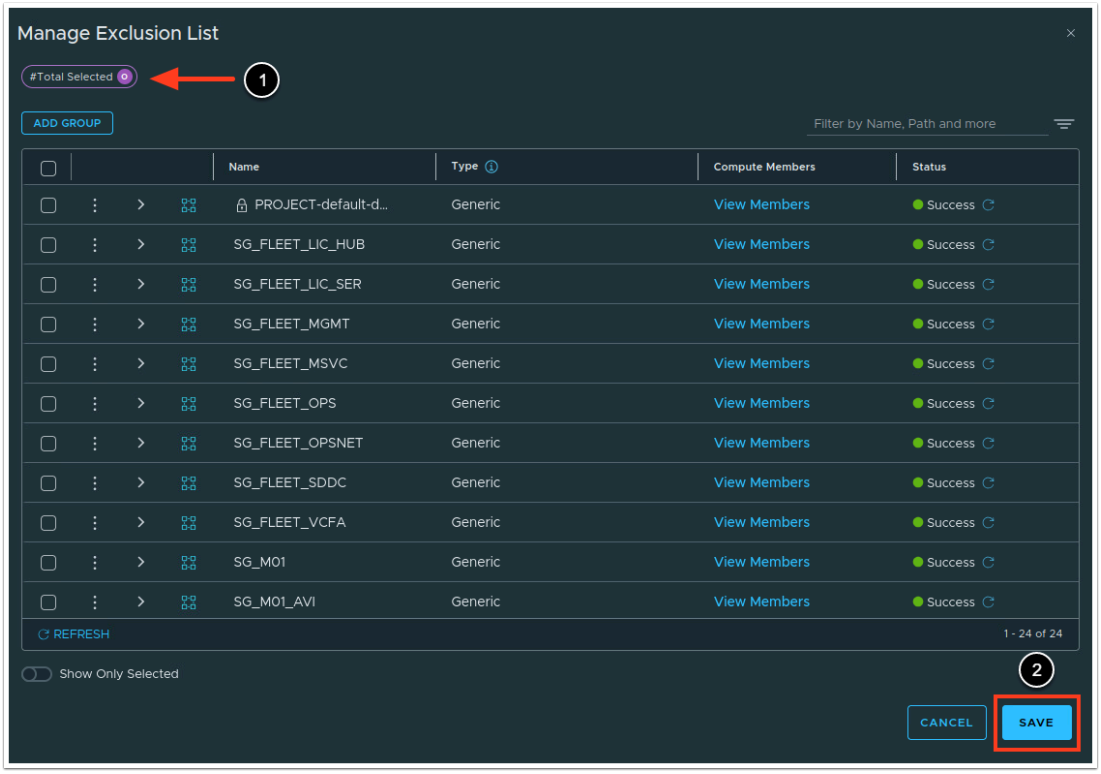
1. Click on **MANAGE EXCLUSION LIST**

Remove the Group



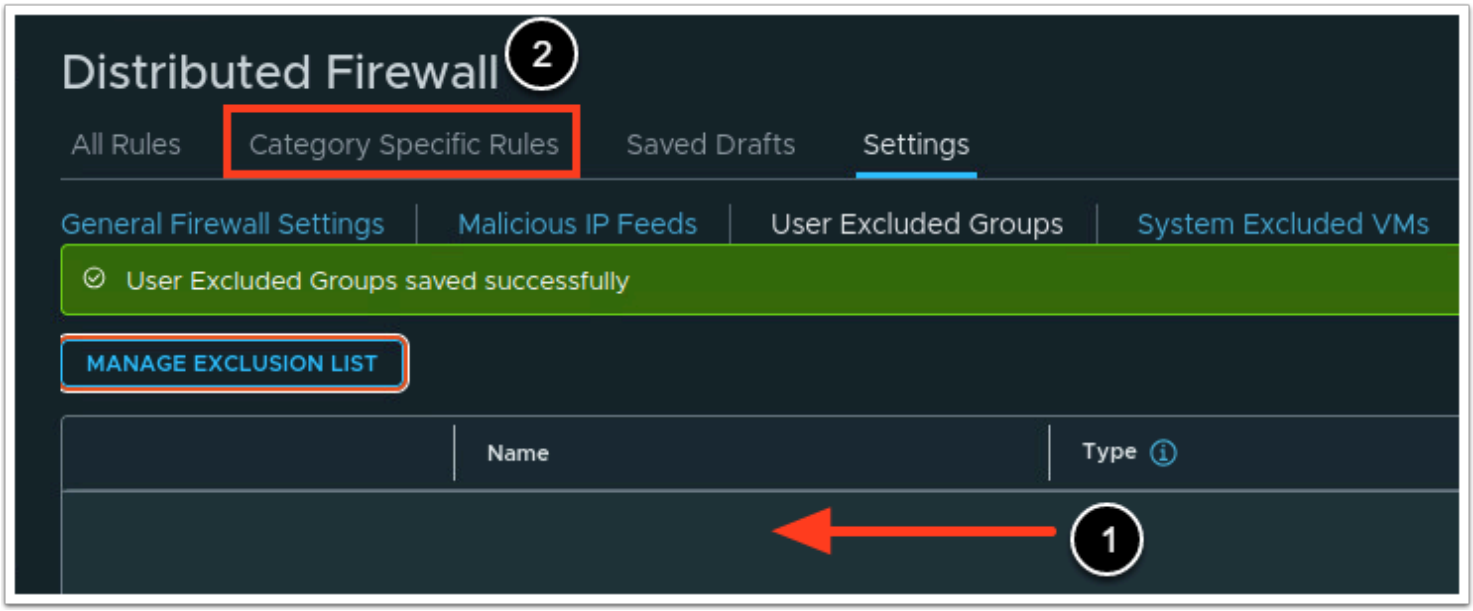
1. Click the toggle for **Show Only Selected**
2. Uncheck the group **VCF-Created-Virtual-Machines**

Verify and Save



1. Verify there are 0 groups selected
2. Click **SAVE**

Return to Category Specific Rules



Now that the VMs have been removed from the Exclusion list you can enable the first policy.

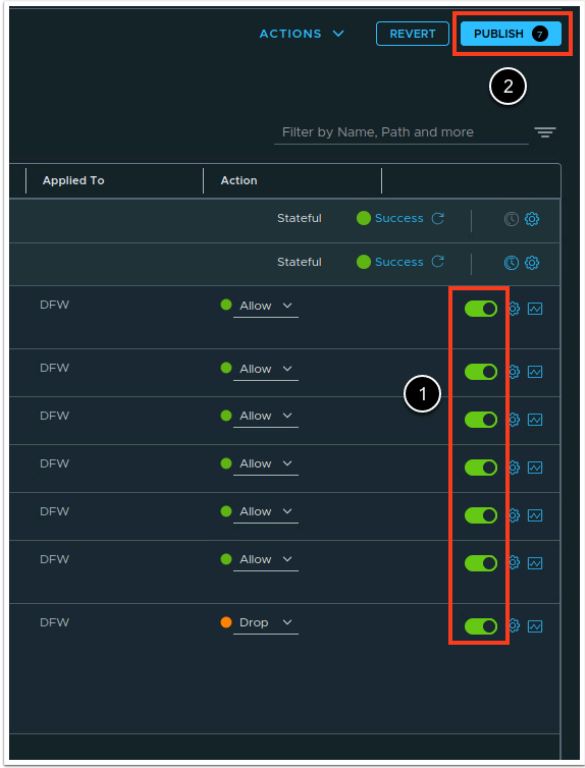
1. Verify that no groups are shown
2. Select **Category Specific Rules** from the Distributed Firewall menu

Navigate to the Infrastructure Section



1. Select the **INFRASTRUCTURE** Section

Enable the rules for the VCF Fleet Policy

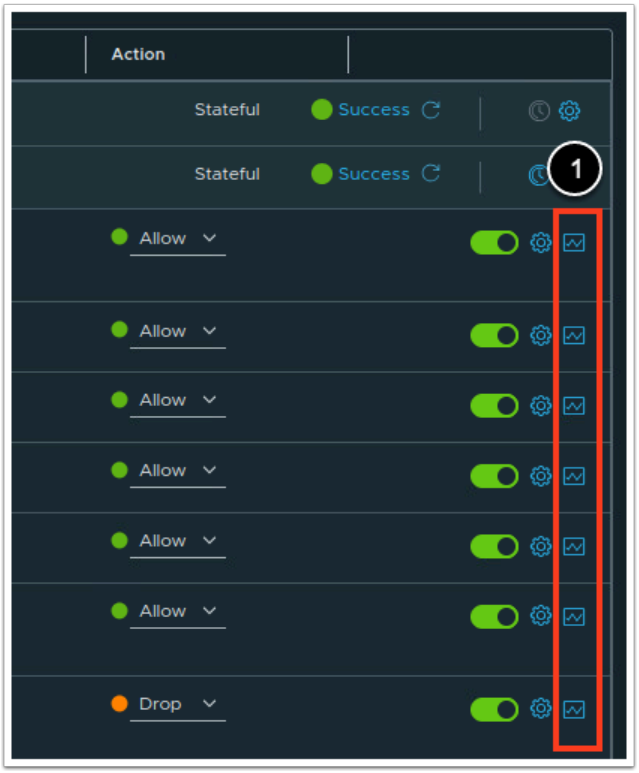


1. Click the toggle for all 7 rules to enable them. They should be green for enabled.
2. Click **Publish**

Task 3: Validate the Infrastructure Policies

In this task you will validate the policies by looking at the counters for the rules that have been created

View the Counters



Each of the rules have a set of counters associated with it to display how many times that rule has been hit. You will use these counters to validate that the rules are being hit.

1. For the first rule (**VCF Fleet to DNS Servers**) click on the **graph** icon on the far right hand side of the rules.

> The counters update every 15 minutes, so it may be some time before you see anything show up in the counter. It is recomended to wait 2 minutes refresh the page and check. If no hits are shown move onto the next module and check the counters during that module's validation task.

Review the Counters



The counters show 3 different pieces of information that can be used.

1. The top set displays the number of times the rule was hit, how much traffic (byte count) and how many packets passed through the rule
2. The middle set displays how frequently it was hit and how that compares to all rules in the system
3. The bottom set shows how many sessions have been processed by the rule and how that compares to all other rules
4. Click the **X** to close out of the window

> Before moving onto the next module take a look at the other policies to see the hit counts on them

Conclusion

Congratulations on completing **Module 1: Initial Steps for Securing the VCF Infrastructure**!

Key Takeaways

- **NSX Policy Review:** Examined existing security configurations and environment policies within NSX Manager.
- **Workload Security:** Enabled Infrastructure security policies to protect the Management Workload Domain.
- **Verification:** Validated policy behavior and verified logging functionality across the environment.

What's Next?
In **Module 2: Configure Environment Rules for the VCF Infrastructure**, you will learn how to define and apply environment-level security rules across your VMware Cloud Foundation deployment.

Ending Your Lab Session

- **Continue to Module 2:** Click **Next** at the bottom of the page or select the next task in the sidebar menu.
- **End Lab Session:** If you are finished with your session, click the **FINISH LAB** button in the lab console.

In this module you reviewed how to configure both Infrastructure and Environment Security Policies for a VCF Deployment.

Module 2 - Configure Environment Rules for the VCF Infrastructure

Introduction

Welcome to Module 2!

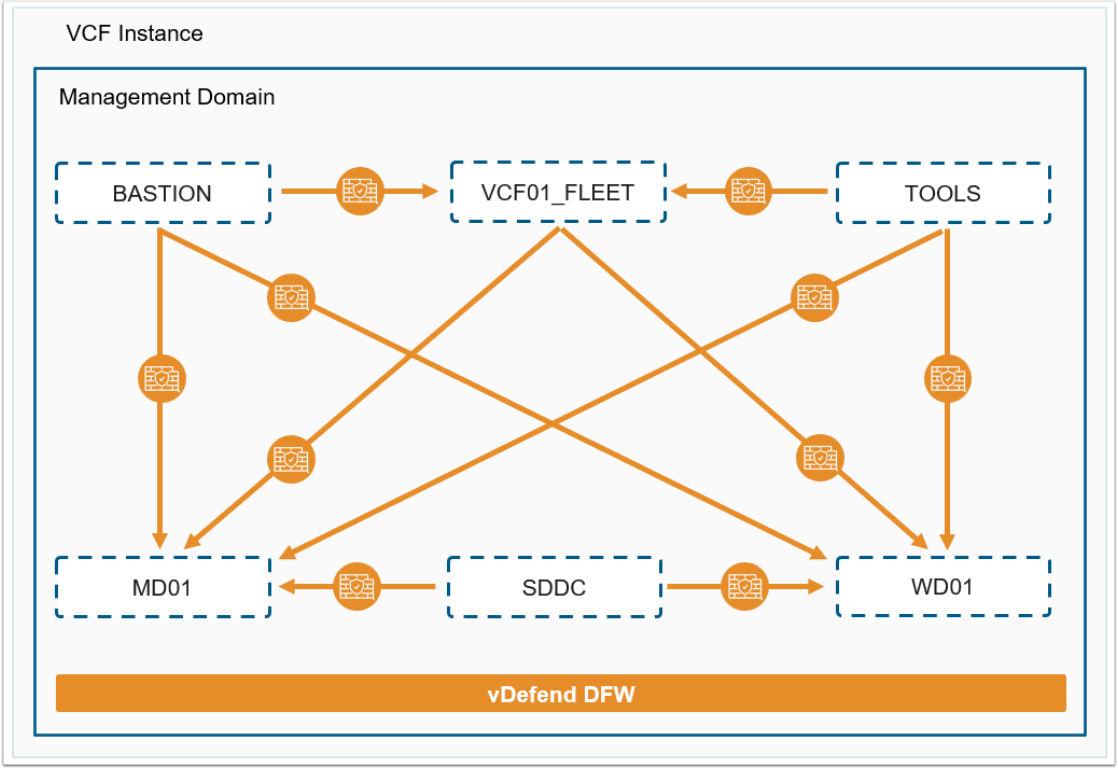
In this module you will review the environment policies for the Management Domain and enable them.

Module Overview

Within this module you will perform the following tasks to enable and monitor distributed firewall policies for the Management Domain

- Task 1: Review the environment security policies for the Management Domain
- Task 2: Enable Environment policies to protect the Management Domain
- Task 3: Validate the Environment Policies

Management Workload Domain Security Strategy



In the previous module you secured the infrastructure protocols that the Management Domain uses for communication. In this module you will be securing the general communication channels between the environments that make up a VCF Management Domain

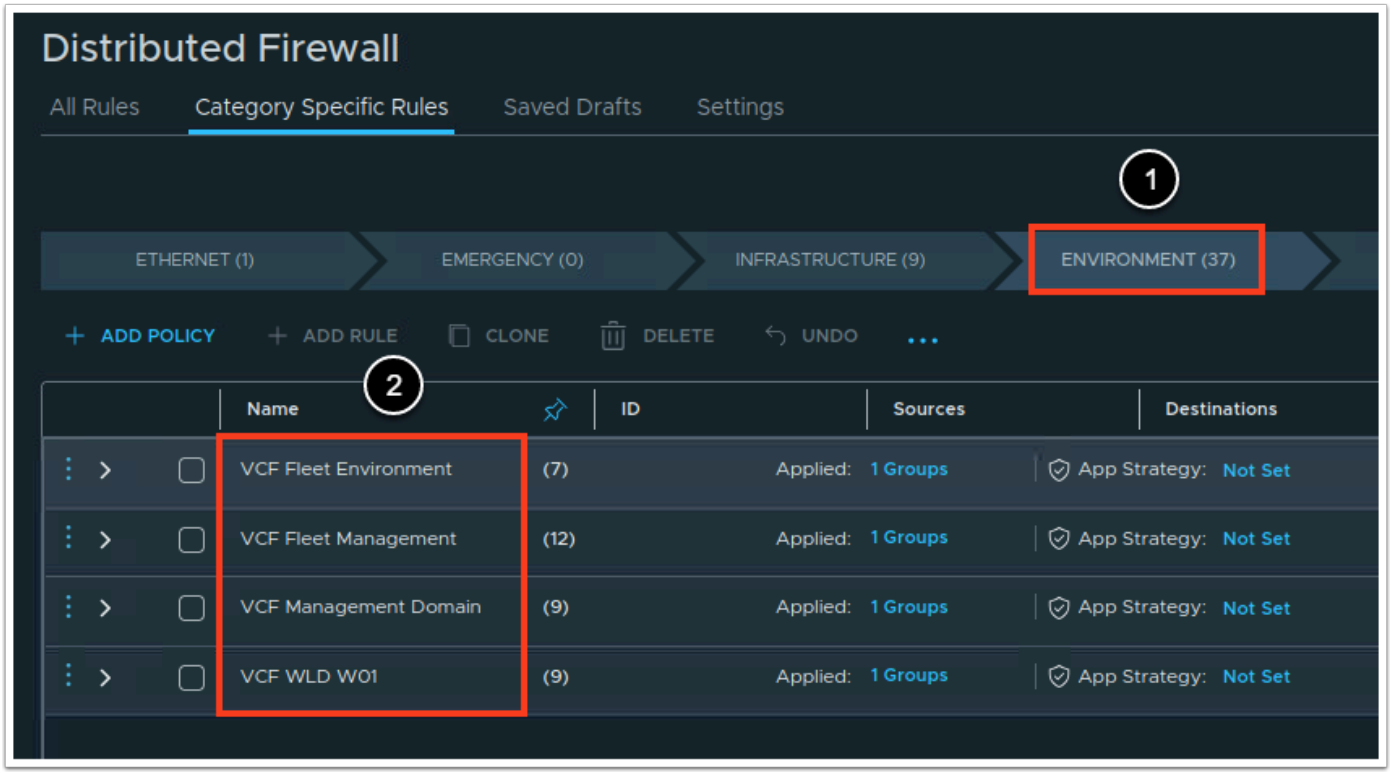
Environment Definitions:

- **VCF01_Fleet:** This is the environment for all virtual machines that are deployed as part of the VCF Fleet that will communicate with each of the different workload domains deployed within the fleet
- **SDDC:** This environment contains the internal components that manage the VCF infrastructure
- **MD01**(Management Domain)**:** This is where core management infrastructure components and workloads are deployed.
- **WD01 (**Workload Domain): This is where non management workloads will be deployed.
- **BASTION:** An environment designated for administrator access, this environment would contain either a collection of jump hosts or the IPs of networks that are allowed to access the Management Domain
- **TOOLS:** An environment designated for automation, external monitoring, and other IT tools that will be used to support the VCF deployment

Task 1: Review the Environment Security Policies for the VCF Deployment

In this task you will review the environment security policies that have been preconfigured for the Management Domain. There will be 4 Policies created and each policy will be reviewed in detail.

Navigate to the Environment Section

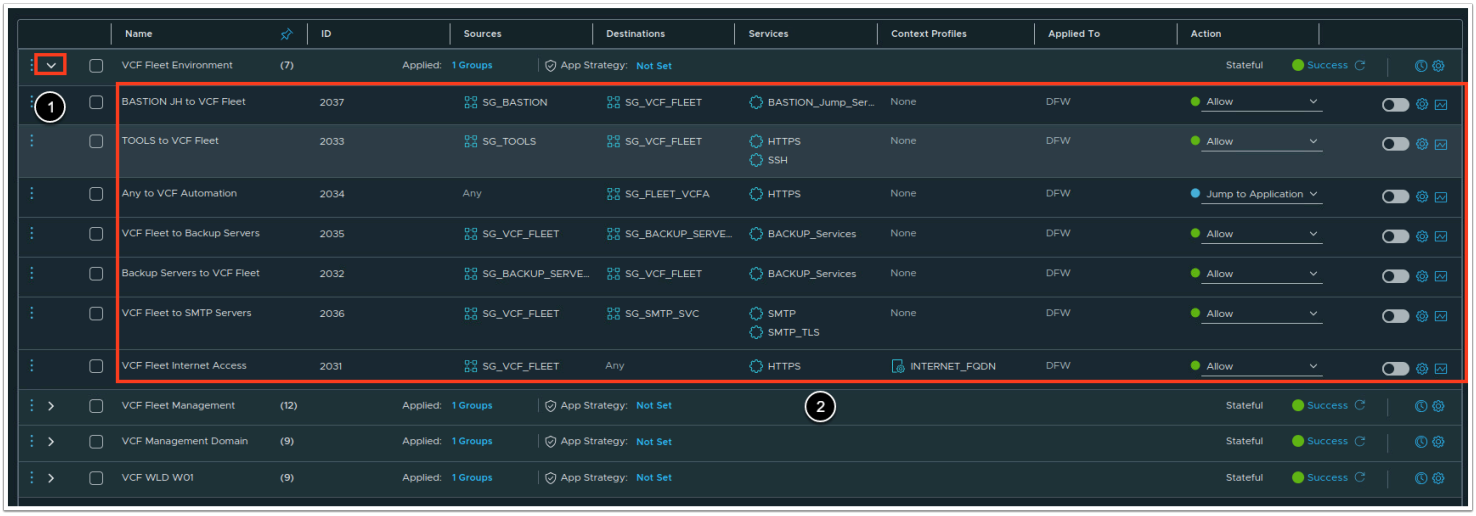


The Environment section contains rules that define how virtual machines can communicate between defined zones or environments. There are 4 policies defined here to communication between zones in the VCF Management Domain.

1. Click the **ENVIRONMENT** section
2. Take note of the 4 Policies
   - The **VCF Fleet Environment** Policy covers communication between the VCF Fleet Members and the external zones such as Bastion and Tools
   - The **VCF Fleet Management** Policy covers communication between the VCF Fleet Members and the Management Domain components of other zones such as the Management Domain and Workload Domain
   - The **VCF Management Domain** Policy covers communication between the VCF Fleet domain and the Management Domain for the purposes of operational functions

- The **VCF WLD W01** Policy covers communication between the VCF Fleet domain and the Workload Domain for the purposes of operational functions

Review the VCF Fleet Environment Policy

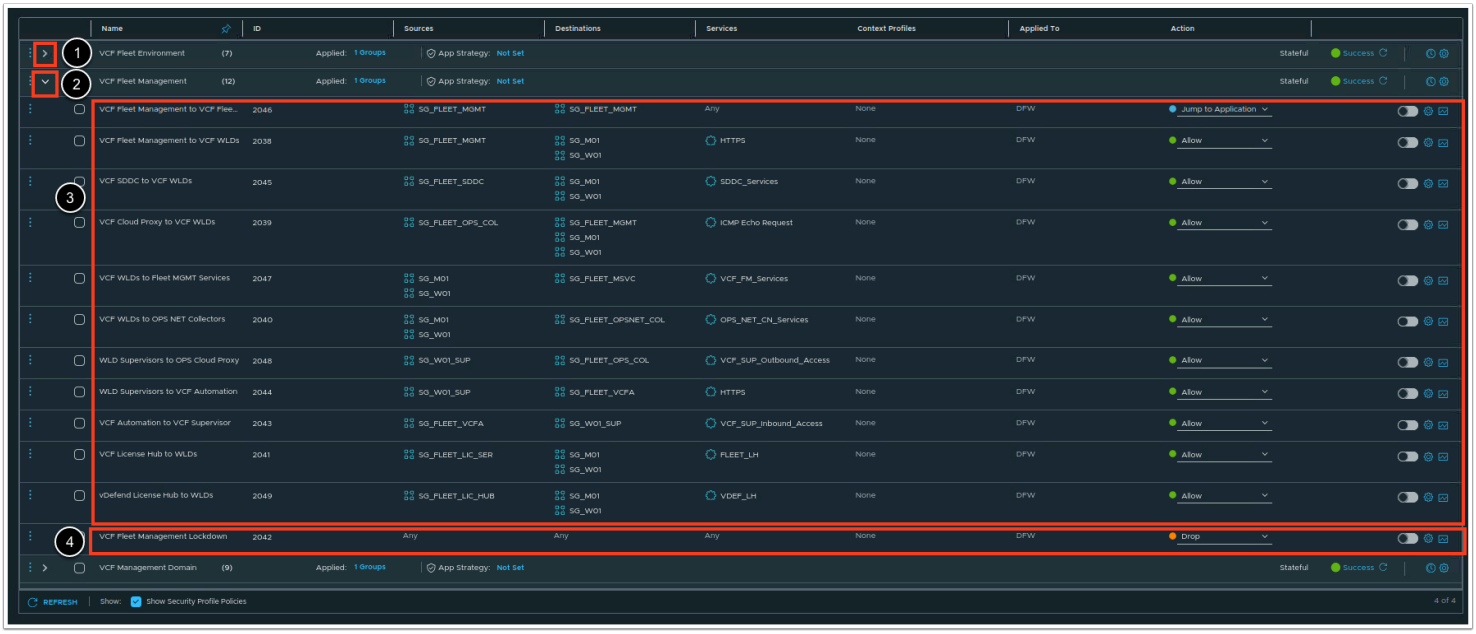


The **VCF Fleet Environment** policy provides a set of policies that define how the Fleet communicates with other zones and how other zones can communicate with it. The policy above allows for Bastion (administration) and Tools (automation and monitoring) access to the environment as well as other management access protocols.

1. Click on the '>' next to **VCF Fleet Environment** to expand the policy
2. There are 7 rules in this policy. Take note of the type of services that are offered.

> You may have noticed that the Any to VCF Automation rule has an action of "Jump to Application". This means that if the conditions for this rule have been met, you will skip the rest of the Environment Section and start evaluating rules again at the beginning of the Application Section. This is a common practice to say you have met condition A but you cannot be allowed until you meet condition B as well.

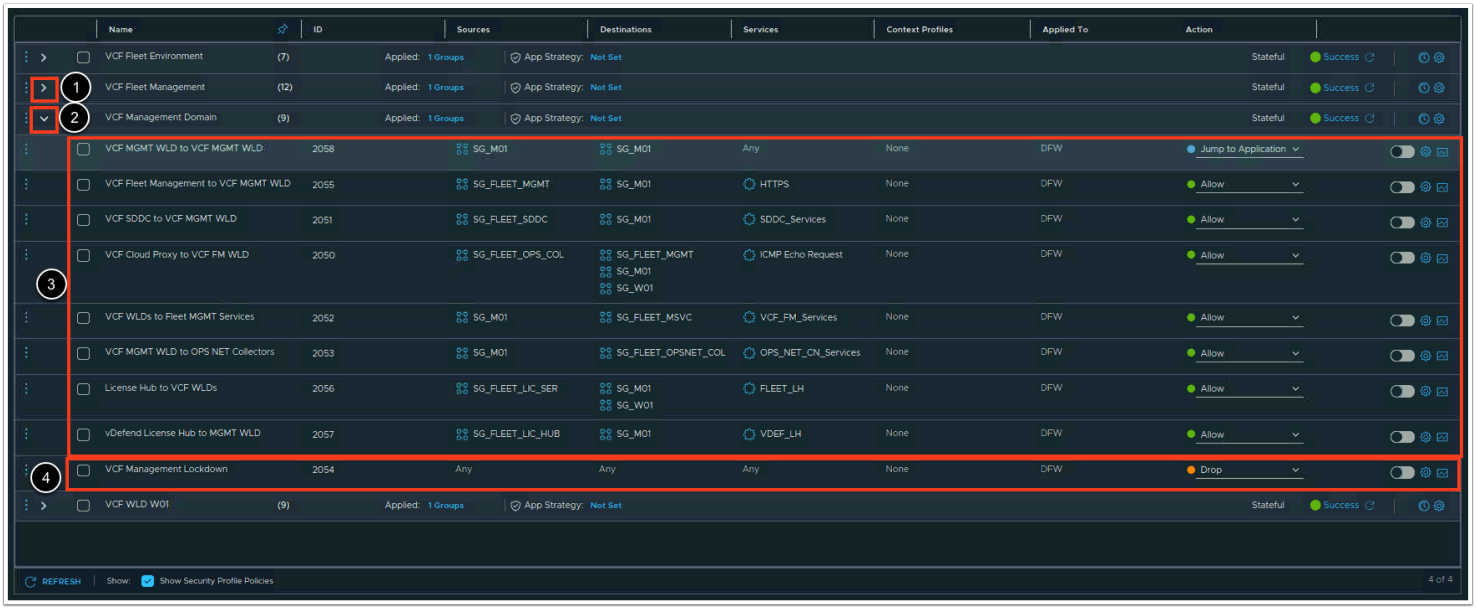
Review the VCF Fleet Management Policy



The **VCF Fleet Management** policy provides a set of policies that define how the Fleet communicates with the workload domains and how the workload domains communicate with the fleet.

1. Click on the '>' next to **VCF Fleet Environment** to collapse the policy
2. Click on the '>' next to **VCF Fleet Management** to expand the policy
3. There are 12 rules in this policy. The first 11 cover how the rules communicate with the Management and Workload Doamins.
   - The last 2 rules allow for communication with the VCF and ANS (SSP) License Hubs.
4. The final rule is a lock down rule to ensure only communication that explicitly allowed by the 11 previous rules is allowed. All other communication is blocked.

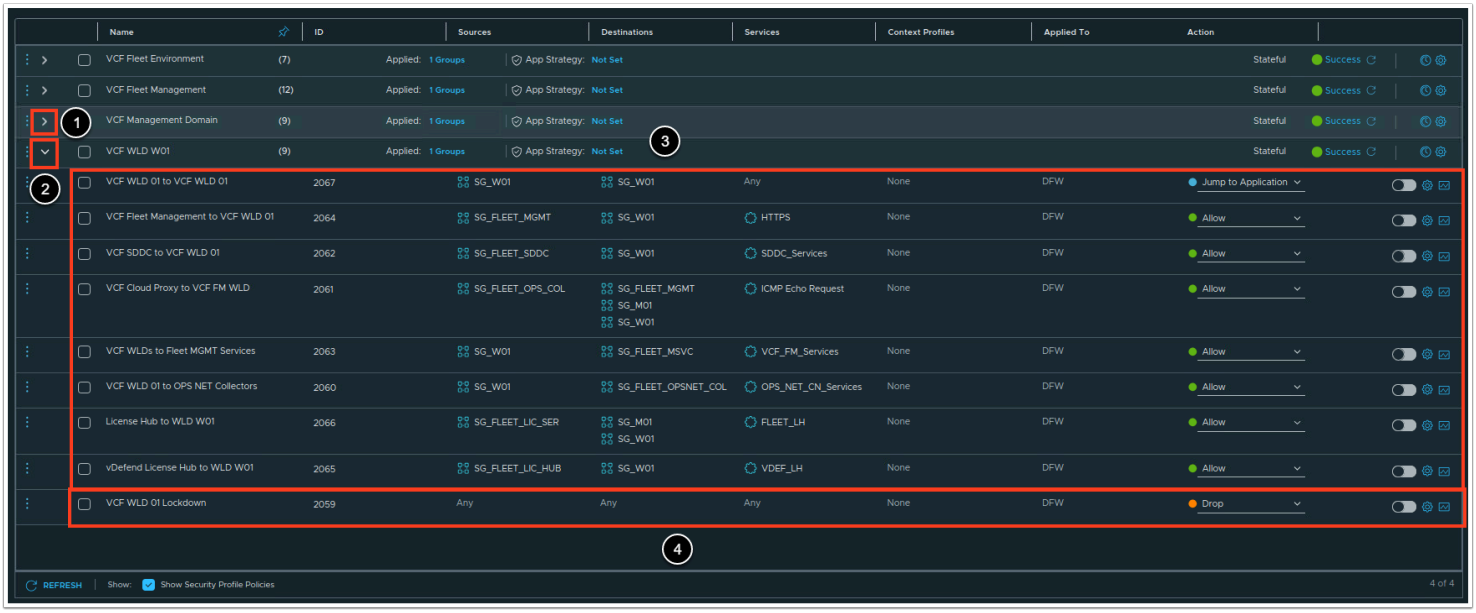
Review the VCF Management Domain Policy



The **VCF Management Domain** policy provides a set of policies that define how the Management Domain communicates with the VCF Fleet Zone.

1. Click on the '>' next to **VCF Fleet Management** to collapse the policy
2. Click on the '>' next to **VCF Management Domain** to expand the policy
3. There are 9 rules in this policy. The first 2 cover how the rules communicate with the Management and Workload Domains.
   - The last 2 rules allow for communication with the VCF and ANS (SSP) License Hubs.
4. The final rule is a lock down rule to ensure only communication that explicitly allowed by the 8 previous rules is allowed. All other communication is blocked.

Review the VCF WLD W01



The **VCF WLD W01** policy provides a set of policies that define how the Workload Domain communicates with the VCF Fleet Zone.

1. Click on the '>' next to **VCF Management Domain** to collapse the policy
2. Click on the '>' next to **VCF WLD W01** to expand the policy
3. There are 9 rules in this policy. The first 2 cover how the rules communicate with the Management and Workload Domains.
   - The last 2 rules allow for communication with the VCF and ANS (SSP) License Hubs.
4. The final rule is a lock down rule to ensure only communication that explicitly allowed by the 8 previous rules is allowed. All other communication is blocked.

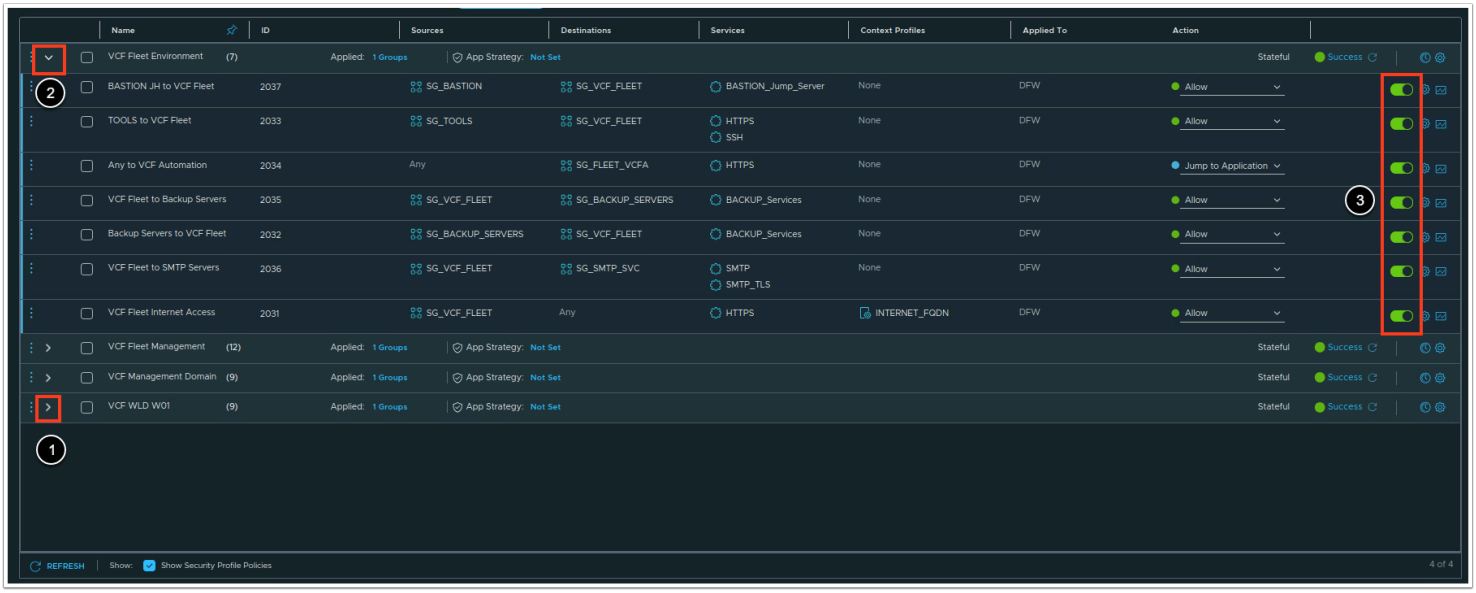
Task 2: Enable Environment policies to protect the VCF Deployment

In this task, you will enable and publish baseline security rules across four VCF policy tiers:

- **VCF Fleet Environment:** Protects shared fleet-wide services and automation tools.
- **VCF Fleet Management:** Secures management services across all domains.
- **VCF Management Domain:** Restricts traffic into the primary management workload domain (mgmt-01a).
- **VCF Workload Domain (W01):** Enforces micro-segmentation for compute workload domains.

*Note: You will enable standard security rules for each policy tier while **intentionally leaving Lockdown rules disabled** to prevent service lockout.*

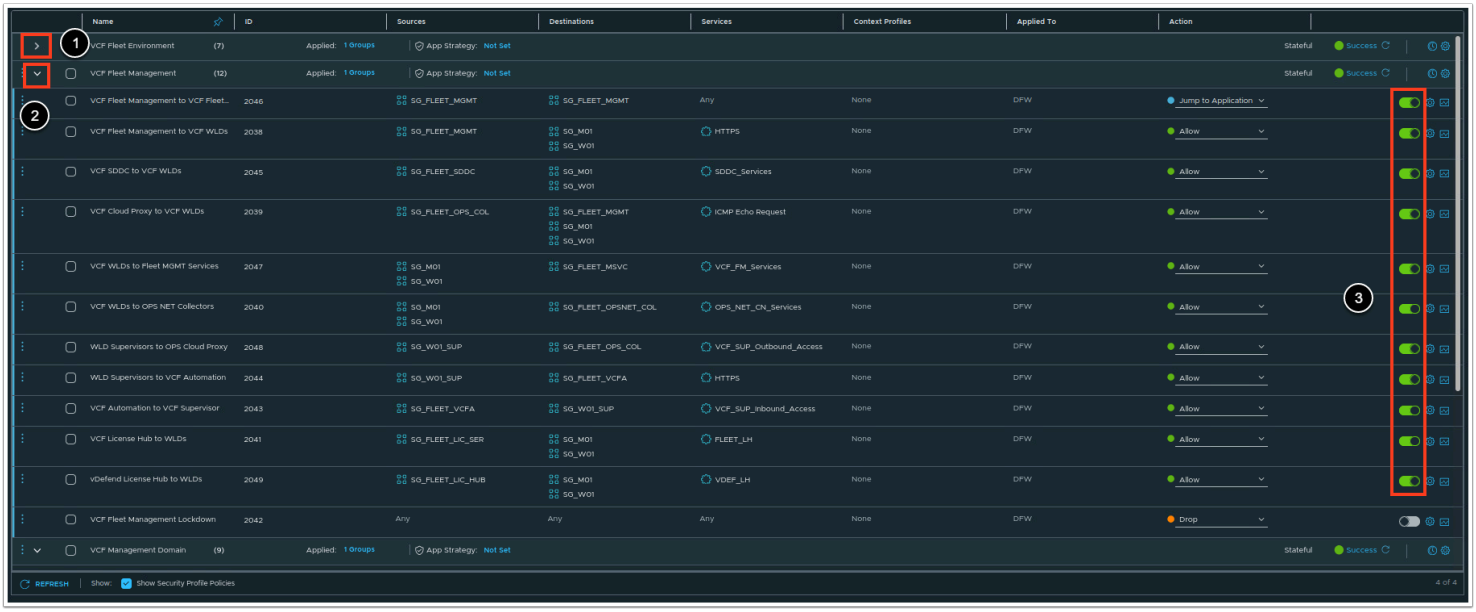
Enable the Rules in the VCF Fleet Environment Policy



1. Click on the '>' next to **VCF WLD W01** to collapse the policy
2. Click on the '>' next to **VCF Fleet Environment** to expand the policy
3. Click the toggle button for each of the 7 rules in the **VCF Fleet Environment** Policy to enable the rules

If less than 7 rules are showing, use the scroll bar on the right to scroll down to ensure all 7 rules will be enabled

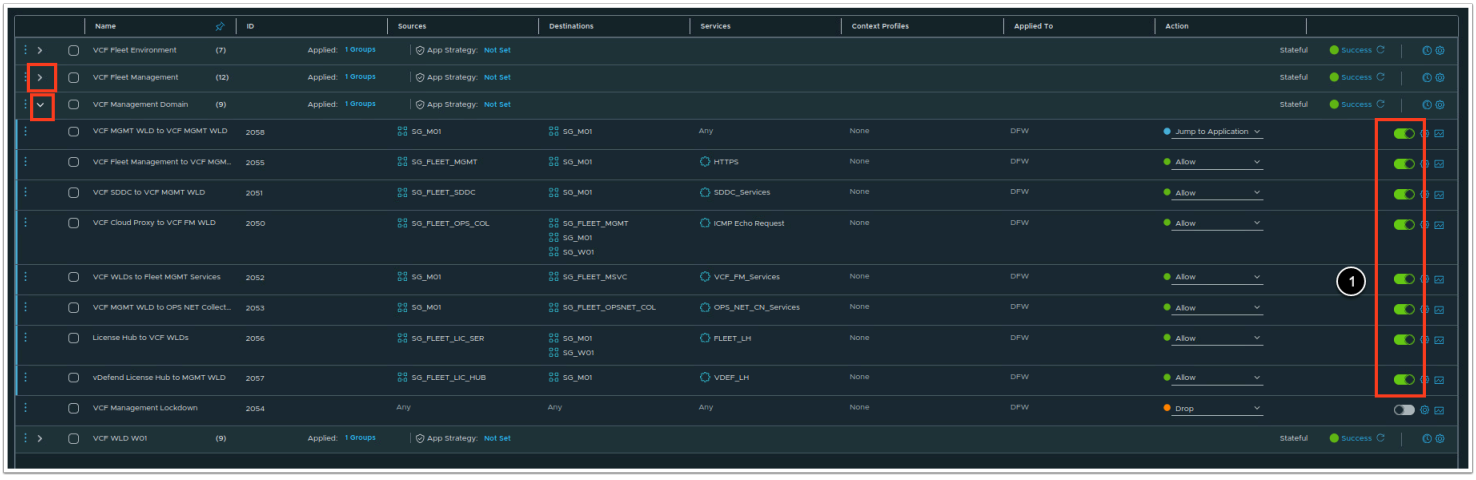
Enable the Rules in the VCF Fleet Management Policy



1. Click on the '>' next to **VCF Fleet Environment** to collapse the policy
2. Click on the '>' next to **VCF Fleet Management** to expand the policy
3. Click the toggle button for the first 11 rules in the **VCF Fleet Management** Policy to enable the rules, **do not enable the Lockdown rule (last rule in the policy).**

If less than 12 rules are showing, use the scroll bar on the right to scroll down to ensure all 12 rules will be enabled

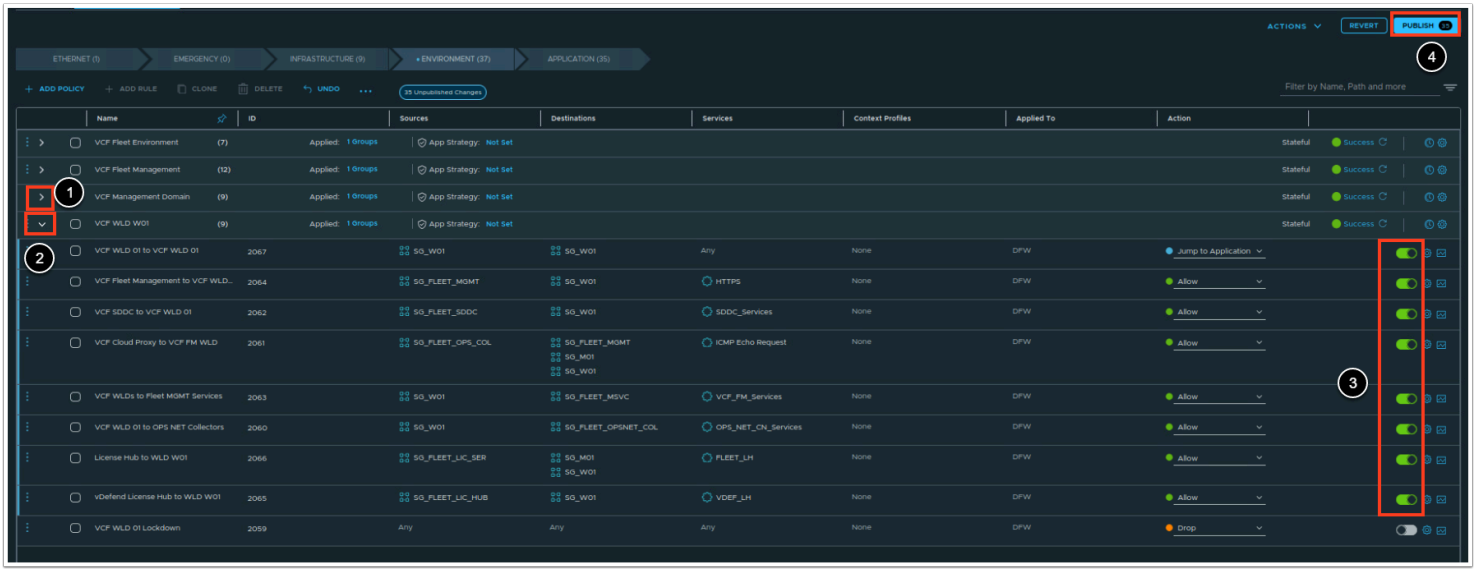
Enable the Rules in the VCF Management Domain Policy



1. Click on the '>' next to **VCF Fleet Management** to collapse the policy
2. Click on the '>' next to **VCF Management Domain** to expand the policy
3. Click the toggle button for the first 8 rules in the **VCF Management Domain** Policy to enable the rules, **do not enable the Lockdown rule (last rule in the policy).**

If less than 8 rules are showing, use the scroll bar on the right to scroll down to ensure all 8 rules will be enabled

Enable the Rules in the VCF Workload Domain Policy and Publish



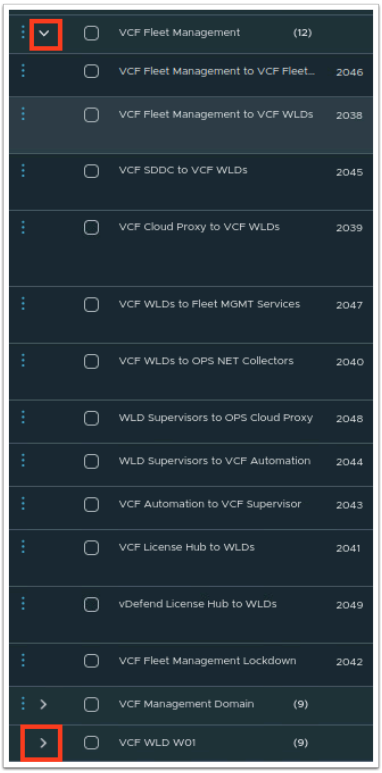
1. Click on the '>' next to **VCF Management Domain** to collapse the policy
2. Click on the '>' next to **VCF WLD W01** to expand the policy
3. Click the toggle button for the first 8 rules in the **VCF WLD W01** Policy to enable the rules, **do not enable the Lockdown rule (last rule in the policy).**
4. Verify there are **35** changes, then click **PUBLISH**

If less than 8 rules are showing, use the scroll bar on the right to scroll down to ensure all 8 rules will be enabled

Task 3: Validate the Environment Policies

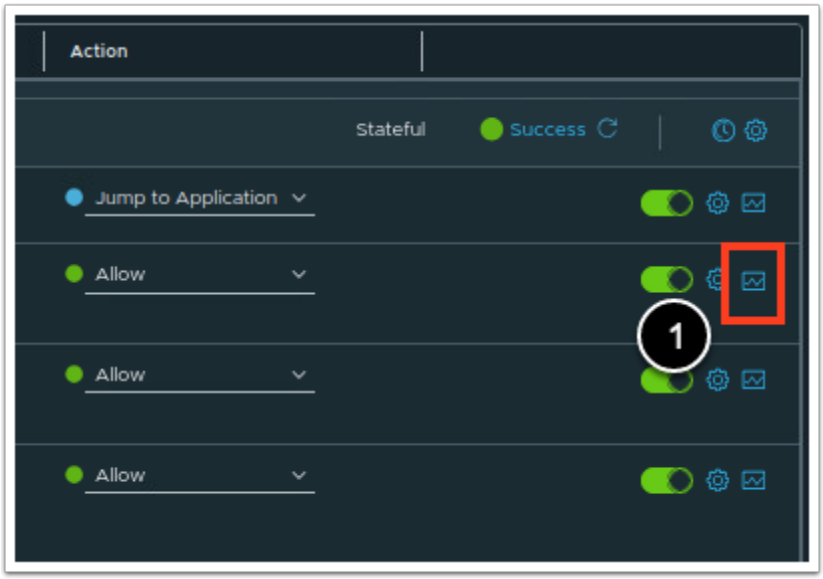
In this task you will validate the policies by looking at the counters for the rules that have been created

Display the VCF Fleet Management Section



1. Click on the '>' next to **VCF WLD W01** to collapse the policy
2. Click on the '>' next to **VCF Fleet Management** to expand the policy

View the Counters

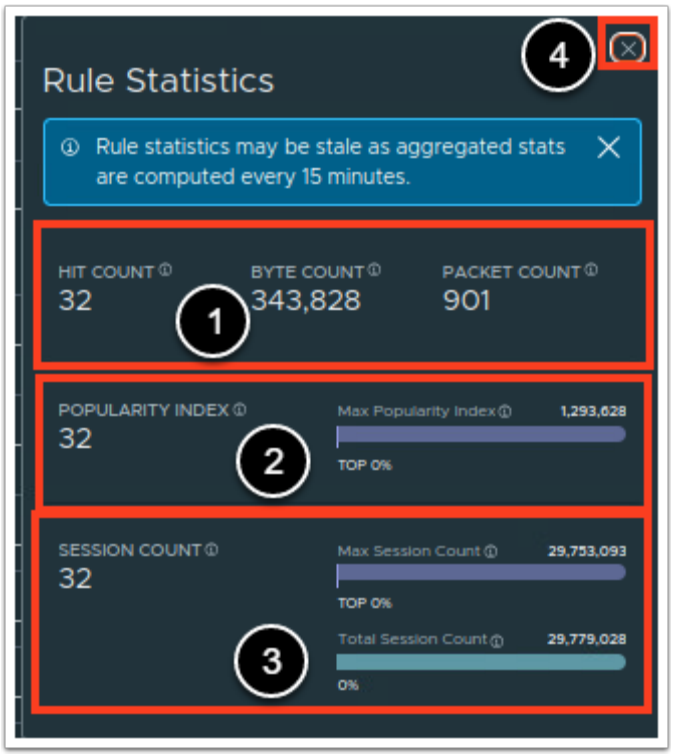


Each of the rules have a set of counters associated with it to display how many times that rule has been hit. You will use these counters to validate that the rules are being hit.

1. For the second rule in the VCF Management policy, rule (**VCF Fleet Management to VCF WLDs**) click on the **graph** icon on the far right hand side of the rules.

> The counters update every 15 minutes, so it may be some time before you see anything show up in the counter. It is recommended to wait 2 minutes refresh the page and check. If no hits are shown move onto the next module and check the counters during that module's validation task.

Review the Counters



The counters show 3 different pieces of information that can be used.

1. The top set displays the number of times the rule was hit, how much traffic (byte count) and how many packets passed through the rule
2. The middle set displays how frequently it was hit and how that compares to all rules in the system
3. The bottom set shows how many sessions have been processed by the rule and how that compares to all other rules
4. Click the **X** to close out of the window

> Before moving onto the next module take a look at the other rules in the other 3 policies to see the hit counts on them

Conclusion

Congratulations on completing Module 2! You have successfully reviewed, enabled, and validated environment-level security policies to harden the VMware Cloud Foundation (VCF) deployment.

Key Accomplishments

- **Policy Review:** Analyzed existing environment security policies to establish baseline protection standards across the VCF infrastructure.
- **Policy Enforcement:** Enabled targeted environment rules to guard core VCF management components against unauthorized lateral access.
- **Operational Validation:** Verified policy enforcement to confirm that infrastructure traffic is appropriately filtered without disrupting management services.

Next Steps
In the next module, you will finalize the security configuration by configuring and enforcing **Application Policies** to protect workload traffic across the environment.

Module 3 - Configure Application Rules for the VCF Infrastructure

Introduction

Welcome to Module 3!

In this final module you will review the application policies for the VCF Infrastructure and enable them.

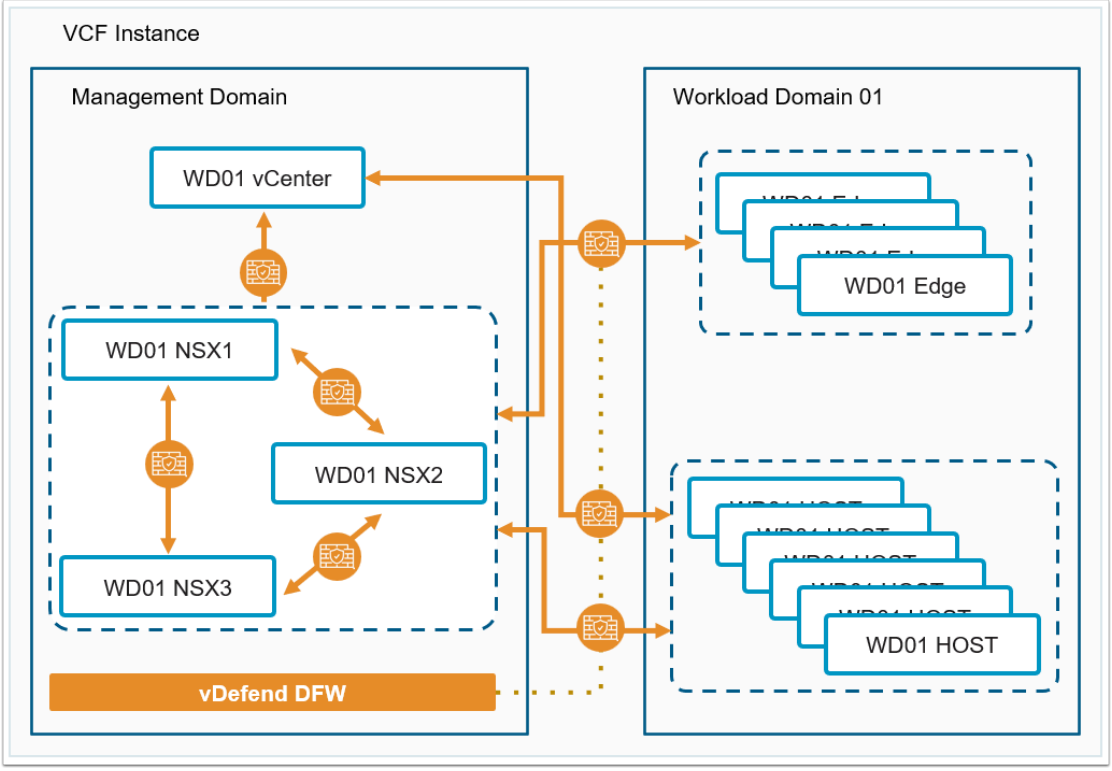
Module Overview

Within this module you will perform the following tasks to enable and monitor distributed firewall policies in the application section for the VCF Infrastructure

• Task 1: Review the application security policies for the VCF Infrastructure

• Task 2: Enable application policies to protect the VCF Infrastructure

• Task 3: Validate the application Policies



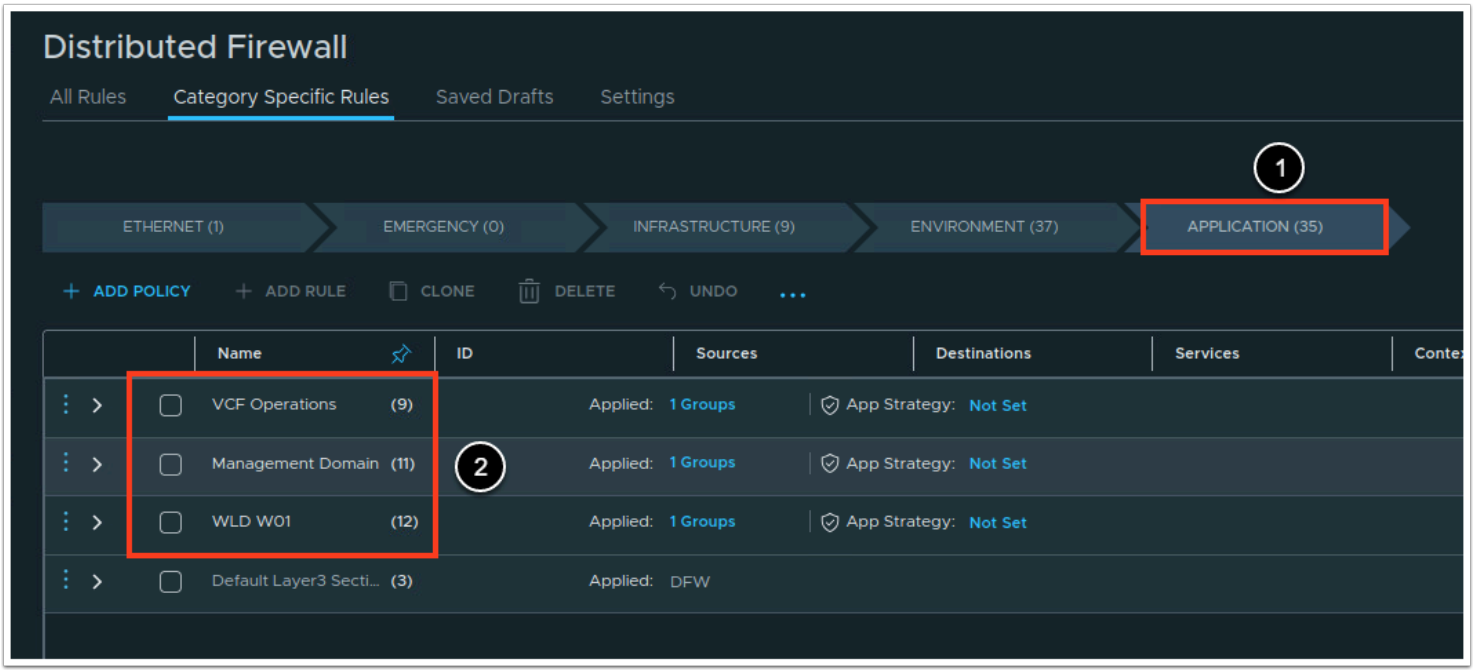
In the previous module you secured the communication between the Fleet Management and the Management and Workload Domains. In this module you will be securing the components of the Workload Domains such as Edges, and Security Service Platform Instances

Each Workload Domain will have very similar characteristics and the policies will not differ significantly between each domain. As with the environment section each Workload Domain will have its own policies and groups.

Task 1: Review the application security policies for the VCF Infrastructure

In this task you will review the environment security policies that have been preconfigured for the Management Domain. There will be 3 Policies created and each policy will be reviewed in detail.

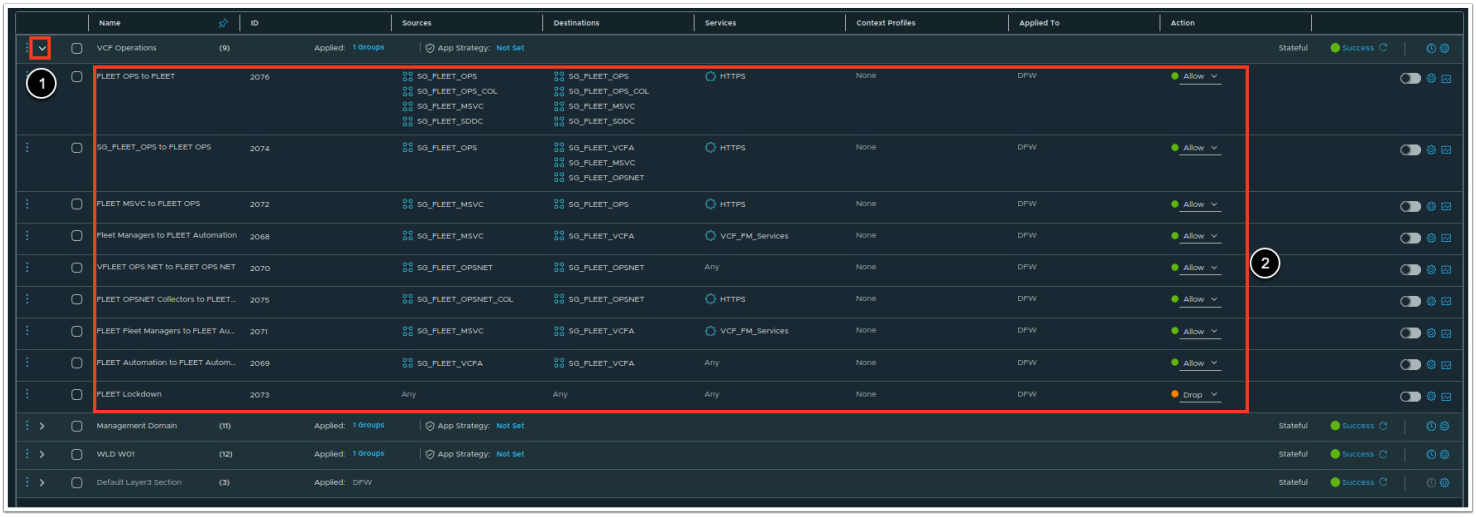
Navigate to the Application Section



The Application section contains rules that define how virtual machines can communicate between defined zones or environments. There are 3 policies defined here to secure communication to infrastructure applications associated with each of the Workload Domains that have been deployed as well as VCF Operations.

1. Click the **APPLICATION** section
2. Take note of the 3 Policies
   - The **VCF Operations** policy covers communication between components of VCF operations.
   - The **Management Domain p**olicy covers communication between the Management Domain vCetner and ESXi Hosts and the other components such as SSP, and Avi
   - The **WLD W01 p**olicy covers communication between the Workload Domain vCetner and ESXi Hosts and the other components such as SSP, and Avi

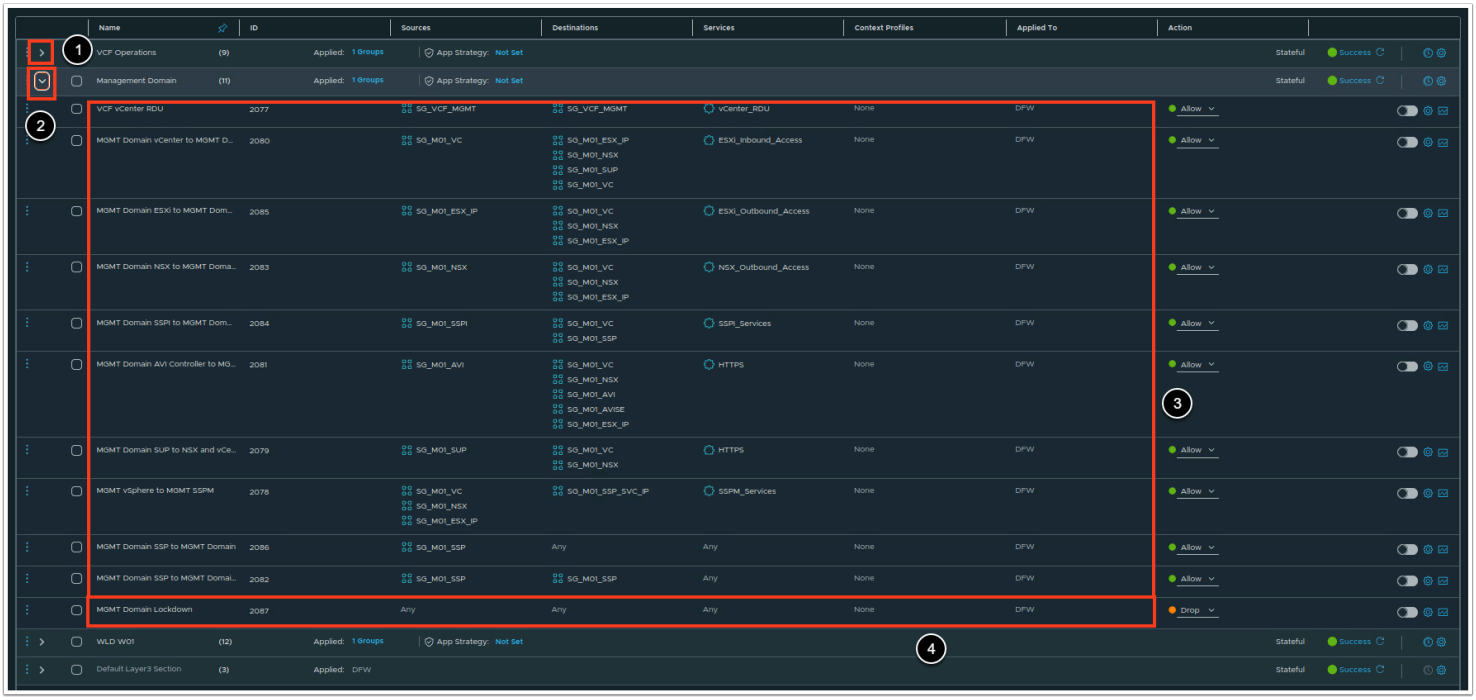
Review the VCF Operations Policy



The **VCF Operations** policy establishes how the components of VCF Operations communicate with each other.

1. Click on the '>' next to **VCF Operations** to expand the policy
2. There are 9 rules in this policy. Take note of how the different components of VCF Operations are expected to communicate with each other

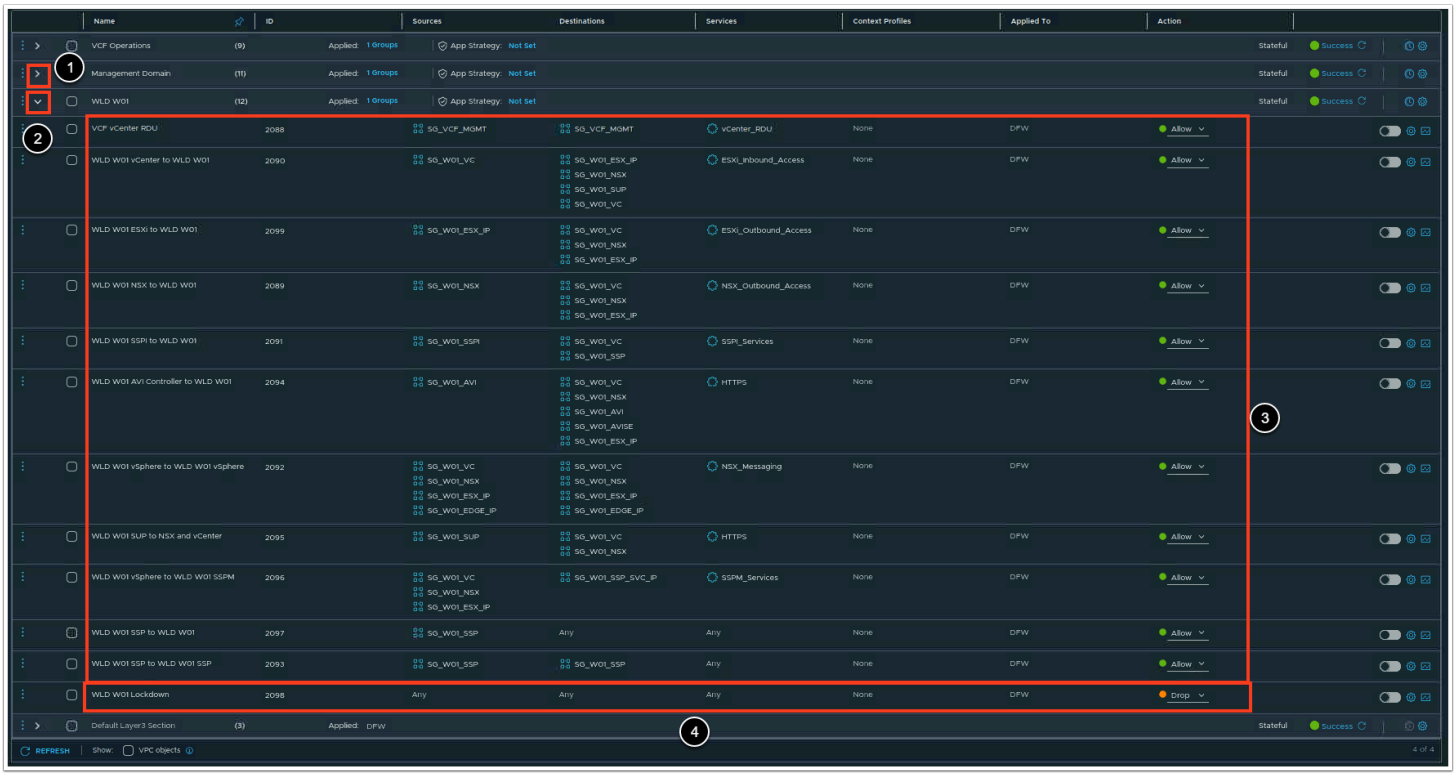
Review the Management Domain Policy



The Management Domain policy is a large policy that support the communication requirements of the vSphere components deployed within the Management domain. These communication paths include vCetner to ESXi, NSX, SSP, and Avi. As well as how other components talk to each other.

1. Click on the '>' next to **VCF Operations** to collapse the policy
2. Click on the '>' next to **Management Domain** to expand the policy
3. There are 11 rules in this policy.
   - The first 8 rules cover the required communication for all of the vSphere components in the environment such as vCenter, Esxi, NSX, VKS, and Avi
   - The next 2 rules cover the required communication for the Security Services Platform.
4. The final rule is a lock down rule to ensure only communication that explicitly allowed by the 11 previous rules is allowed. All other communication is blocked.

Review the VCF Workload Domain Policy



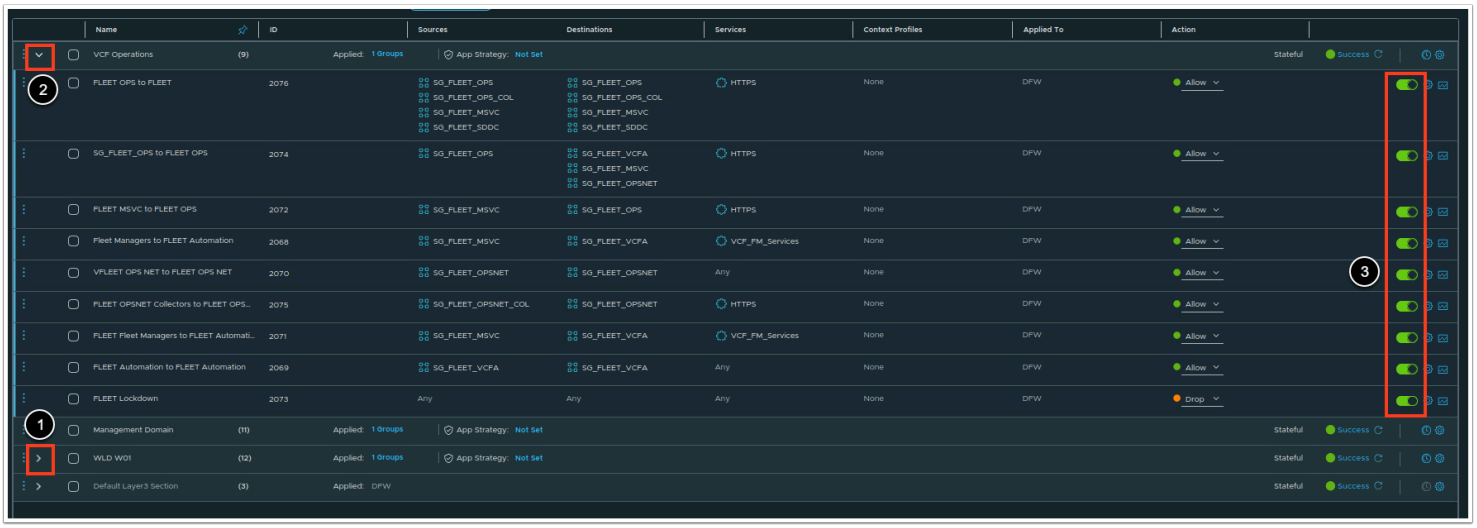
The WLD W01 policy is a large policy that support the communication requirements of the vSphere components deployed within the Workload domain. These communication paths include vCetner to ESXi, NSX, SSP, and Avi. As well as how other components talk to each other. An additional policy is created to support NSX communication with the Workload Domain that is not covered by an Environment rule like the Management Domain NSX manager is.

1. Click on the '>' next to **Management Domain** to collapse the policy
2. Click on the '>' next to **WLD W01** to expand the policy
3. There are 11 rules in this policy.
   - The first 9 rules cover the required communication for all of the vSphere components in the environment such as vCenter, Esxi, NSX, VKS, and Avi
   - The next 2 rules cover the required communication for the Security Services Platform.
4. The final rule is a lock down rule to ensure only communication that explicitly allowed by the 11 previous rules is allowed. All other communication is blocked.

Task 2: Enable application policies to protect the VCF Infrastructure

The next task will be to enable the Application Policies and its associated rules for the VCF Infrastructure

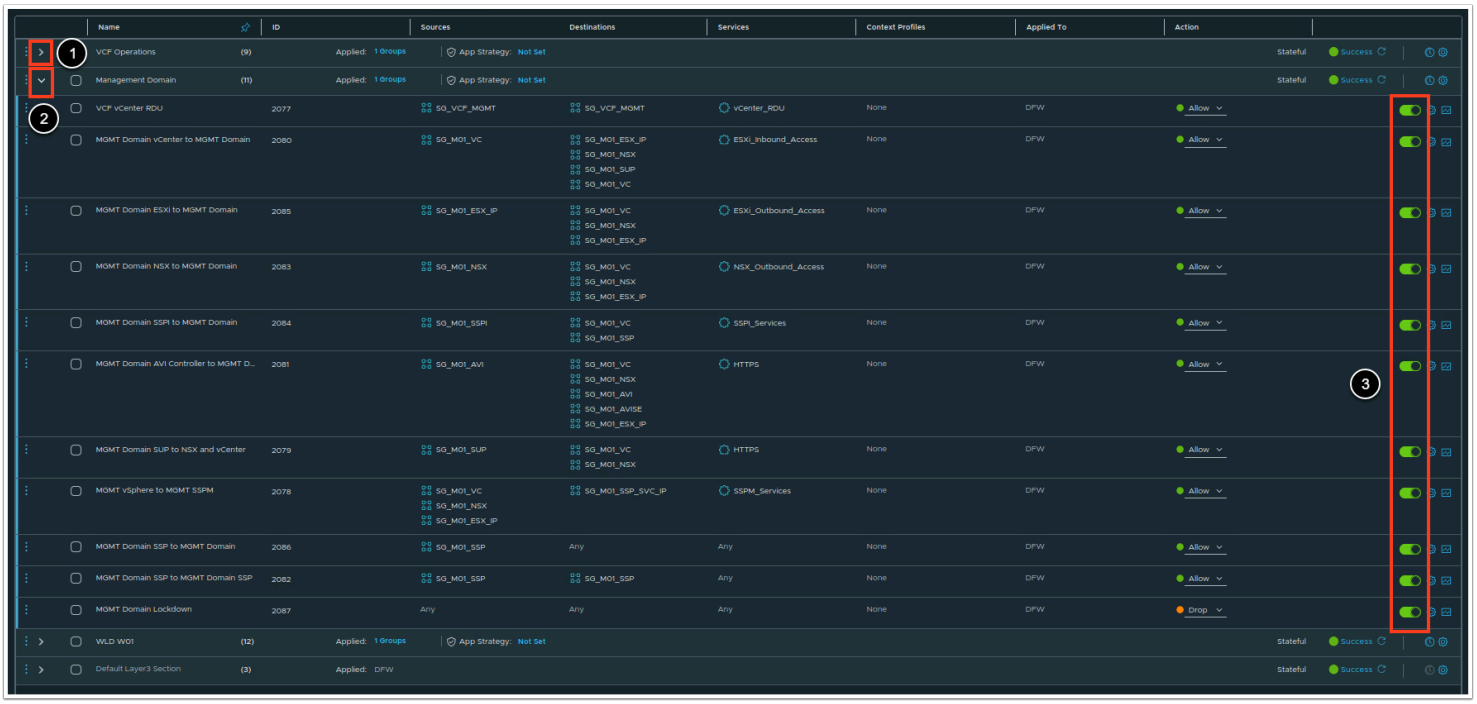
Enable the Rules in the VCF Operations Policy



1. Click on the '>' next to **WLD W01** to collapse the policy
2. Click on the '>' next to **VCFOperations** to expand the policy
3. Click the toggle button for each of the 9 rules in the **VCF Operations** policy to enable the rules

If less than 7 rules are showing, use the scroll bar on the right to scroll down to ensure all 7 rules will be enabled

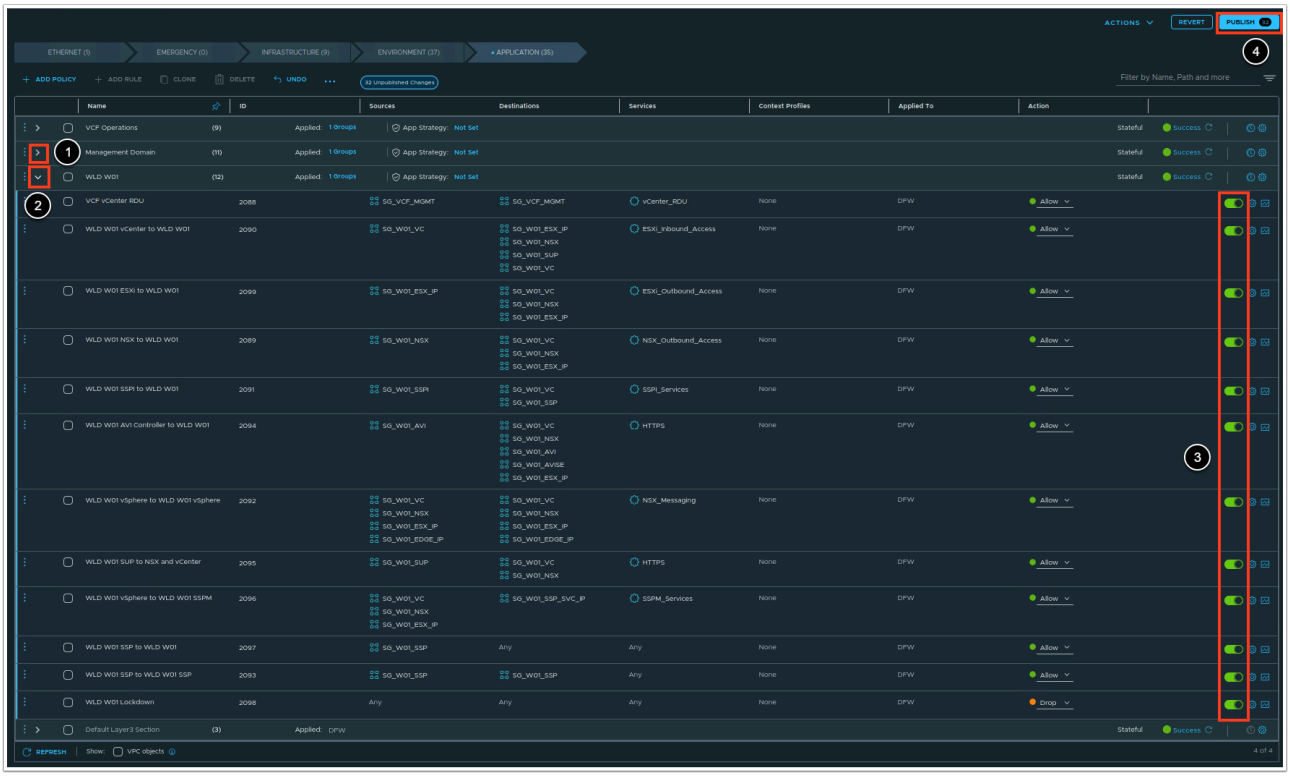
Enable the Rules in the Management Domain Policy



1. Click on the '>' next to **VCF Operations** to collapse the policy
2. Click on the '>' next to **Management Domain** to expand the policy
3. Click the toggle button for the first 11 rules in the **Management Domain** policy to enable the rules

If less than 11 rules are showing, use the scroll bar on the right to scroll down to ensure all 11 rules will be shown

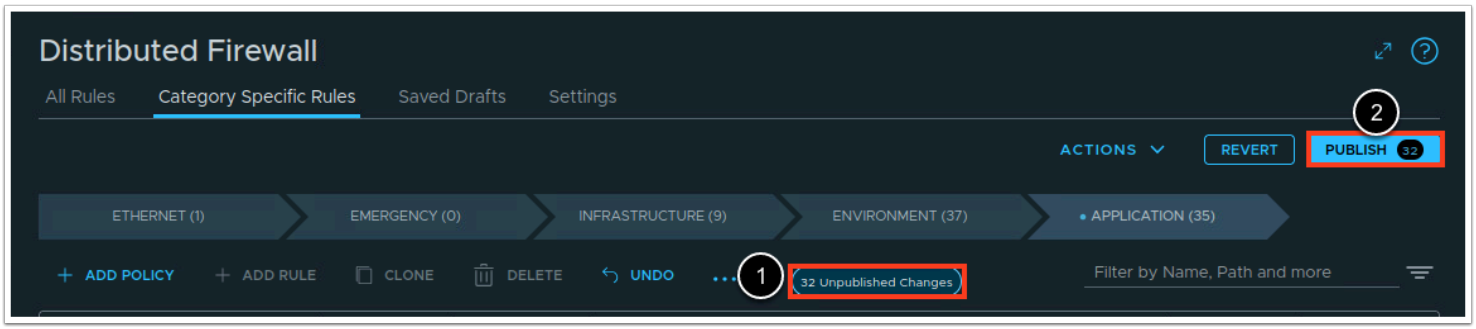
Enable the Rules in the WLD W01 Policy



1. Click on the '>' next to **Management Domain** to collapse the policy
2. Click on the '>' next to **WLD 01** to expand the policy
3. Click the toggle button for the first 11 rules in the **WLD W01** policy to enable the rules

If less than 12 rules are showing, use the scroll bar on the right to scroll down to ensure all 12 rules will be shown

Publish the rules



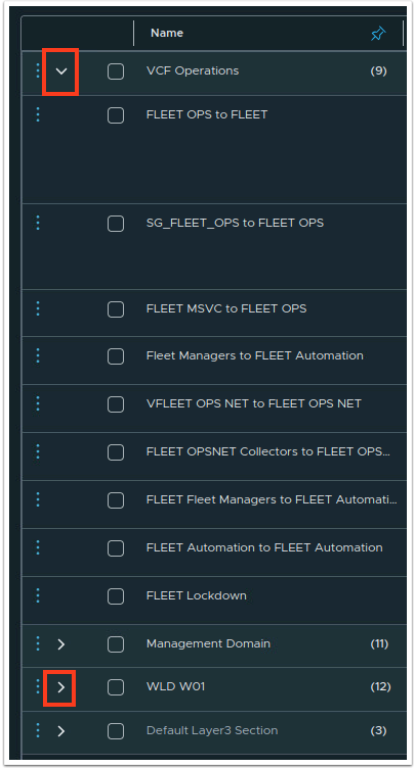
1. Ensure that there are **32** changes that will be published.

2. click **PUBLISH**

Task 3: Validate the application Policies

In this task you will validate the policies by looking at the counters for the rules that have been created

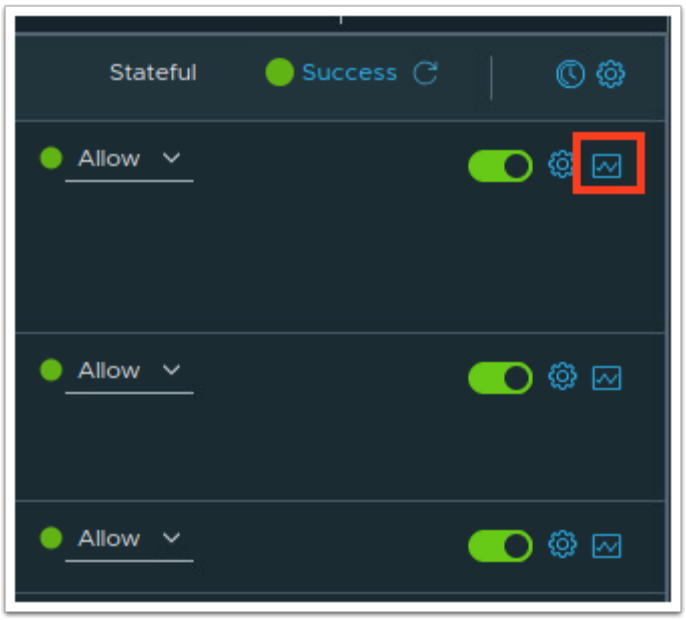
Display the VCF Operations Policy



The most talkative rule will be the first rule on the VCF Operations Policy. You will be using that to inspect the counters.

1. Click on the '>' next to **VCF WLD W01** to collapse the policy
2. Click on the '>' next to **VCF Fleet Management** to expand the policy

View the Counters

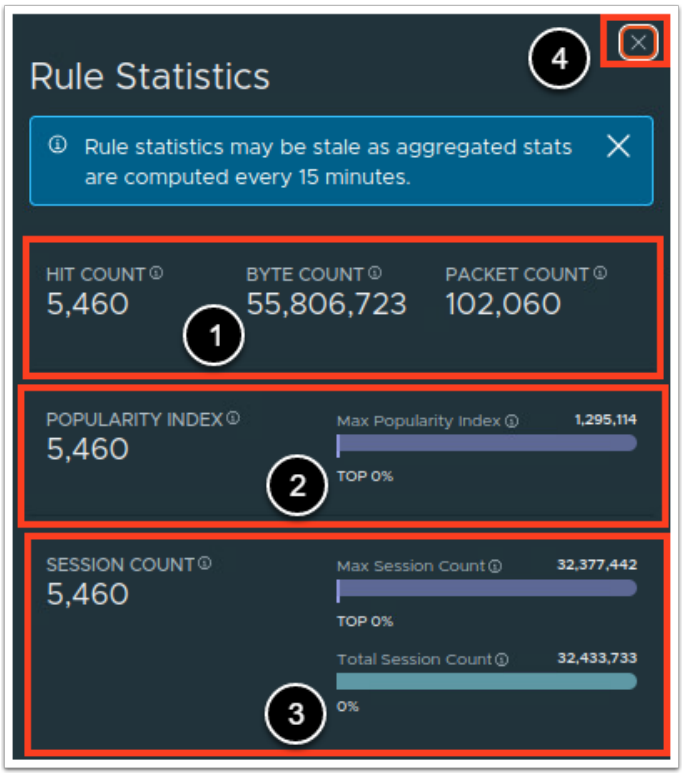


Each of the rules have a set of counters associated with it to display how many times that rule has been hit. You will use these counters to validate that the rules are being hit.

1. For the first rule in the VCF Operations policy, rule (FLEET OPS to FLEET) click on the **graph** icon on the far right hand side of the rules.

> The counters update every 15 minutes, so it may be some time before you see anything show up in the counter. It is recommended to wait 2 minutes refresh the page and check.

Review the Counters



The counters show 3 different pieces of information that can be used.

1. The top set displays the number of times the rule was hit, how much traffic (byte count) and how many packets passed through the rule
2. The middle set displays how frequently it was hit and how that compares to all rules in the system
3. The bottom set shows how many sessions have been processed by the rule and how that compares to all other rules
4. Click the **X** to close out of the window

> This is the last task you will complete. Feel free to spend some time to familiarize yourself with the policies and how they function as well as look to see which ones are being hit in the lab and which ones aren't

Conclusion

Congratulations on completing Module 3! You have successfully configured, enforced, and validated application-level security policies to harden the VMware Cloud Foundation (VCF) environment.

Key Accomplishments

- **Security Audit:** Reviewed existing application security policies to establish baseline communication rules for VCF components.
- **Policy Enforcement:** Enabled targeted application rules using vDefend to block unauthorized lateral movement across east-west traffic.
- **Validation:** Verified rule enforcement to ensure critical management services and workloads remain protected without disrupting legitimate operational traffic.

Next Steps & Recommended Reading

To further extend your understanding of VCF security design and micro-segmentation best practices, explore these key resources:

- **vDefend Blueprints:** Explore lateral security architectures in **Introducing VMware vDefend Lateral Security Design Blueprints for VCF 9.1**
- **Implementation Guides:** Review The VMware Validated Solution to Secure the VCF Infrastructure for production deployment standards.

Ending the Lab

You have reached the end of Module 3 and completed all lab requirements. Click **End Lab** in the top navigation panel of your lab interface to release your assigned virtual resources.

End of Lab Manual



vmware®

by Broadcom

Copyright © 2025 Broadcom. All rights reserved.
The term “Broadcom” refers to Broadcom Inc. and/or its subsidiaries. For more information, go to www.broadcom.com. All trademarks, trade names, service marks, and logos referenced herein belong to their respective companies. Broadcom reserves the right to make changes without further notice to any products or data herein to improve reliability, function, or design.

Information furnished by Broadcom is believed to be accurate and reliable. However, Broadcom does not assume any liability arising out of the application or use of this information, nor the application or use of any product or circuit described herein, neither does it convey any license under its patent rights nor the rights of others.
Item No: 51227-vcf-wp-hands-on-labs-manual-2025, Jan-25